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ASPECT-BASED SENTIMENT ANALYSIS OF
AIRLINE REVIEWS: A DOMAIN-AWARE NLP
STUDY

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Abstract

Airline passenger satisfaction is rarely the result of a single service experience or a single emotional judgement. A review may describe a delayed departure, a helpful crew, an uncomfortable seat, and a disappointing perception of value within the same narrative. For this reason, document-level sentiment analysis is not sufficient to explain how passengers evaluate airline service quality. This thesis addresses this limitation by developing a domain-aware Aspect-Based Sentiment Analysis (ABSA) framework for large-scale airline passenger reviews.

The study is based on a corpus of 144,934 passenger reviews covering 578 airlines and flights from April 2012 to March 2026, collected from airlinequality.com. The proposed pipeline combines aspect extraction, aspect sentiment classification, rating–aspect alignment, predictive modelling, and contextual analysis. Aspect extraction is performed through BIO sequence tagging and identifies 717,904 aspect spans across five airline-specific service dimensions: Punctuality, Cabin Comfort, Cabin Crew Service, Food & Beverages, and Value For Money. The high review coverage of 95.8% confirms that the aspect schema captures the main service dimensions discussed by passengers.

For aspect sentiment classification, the thesis compares a lexicon baseline, a TF-IDF model with Logistic Regression, and a fine-tuned DistilBERT model trained under rating-derived silver-label supervision. The TF-IDF

model achieves the strongest overall performance, with a macro-F1 of 0.608, improving substantially over the lexicon baseline and outperforming DistilBERT on four of five aspects. DistilBERT reaches an overall macro-F1 of 0.574 and struggles especially with the neutral class. This result suggests that, in this setting, the main bottleneck is not model capacity but the quality and ambiguity of silver-label supervision.

The empirical findings show that Value For Money is the strongest driver of overall passenger satisfaction. It emerges as dominant across Pearson correlation with the overall rating, Ridge regression feature importance, and AOPE polarity analysis. By contrast, Punctuality is the only aspect with negative mean AOPE polarity, indicating that delays operate as a non-compensable damage factor: once reliability fails, other service dimensions are less able to repair the passenger’s overall evaluation.

The thesis also shows that satisfaction is strongly shaped by context. Carrier type, cabin class, route length, aircraft type, traveller segment, delay mentions, weather attribution, seasonal demand, and major aviation events all influence how passengers interpret service quality. Weather-related language partly softens blame when disruptions are perceived as externally caused, while peak-season analysis reveals systematic satisfaction penalties, especially in the Northern Hemisphere and among Premium Economy passengers. Difference-in-Differences analysis further shows that major aviation events cannot be interpreted through raw before-and-after comparisons alone, because seasonal trends can change the apparent direction of the effect.

Overall, the thesis contributes both methodologically and managerially. Methodologically, it demonstrates that a modular, domain-aware ABSA pipeline can extract meaningful aspect-level and contextual evidence from large-scale airline reviews. Managerially, it shows that passenger satisfaction is governed

by perceived fairness, operational reliability, expectation management, and contextual attribution. The results confirm that airline reviews should not be treated as simple positive or negative opinions, but as structured narratives through which passengers explain what mattered, what failed, and under which conditions service failures became most damaging.

Keywords: Aspect-Based Sentiment Analysis, airline reviews, passenger satisfaction, TF-IDF, DistilBERT, BIO tagging, silver labels, weather attribution, seasonality, causal inference.

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Chapter 1

Introduction

Customer reviews have become one of the most important sources of information for understanding how people evaluate service quality. In many industries, customers no longer express satisfaction only through formal surveys or direct complaints. They also describe their experiences publicly, in their own words, on digital review platforms. These reviews are especially valuable because they combine evaluation, emotion, explanation, and context. They do not simply state whether an experience was good or bad; they often explain what happened, which part of the service mattered most, and why the customer judged the experience in a certain way.

This is particularly relevant in the airline industry. Air travel is not a single service moment, but a sequence of interconnected experiences. A passenger may interact with booking systems, airport staff, boarding procedures, cabin crew, onboard services, aircraft facilities, baggage processes, and disruption-management systems within the same journey. Each of these moments can shape the final perception of the airline. A passenger may praise the professionalism of the crew while criticising a delay, complain about seat comfort while still considering the ticket good value, or give a low overall rating because of poor communication during disruption even if the flight

itself was acceptable. For this reason, airline satisfaction is inherently multi-dimensional.

Traditional sentiment analysis is limited in this context because it usually assigns one overall polarity to an entire document. This can be useful when the objective is to classify a review as generally positive or negative, but it is not sufficient when the review contains several opinions about different parts of the service. A single airline review may contain positive, negative, and neutral evaluations at the same time. Treating the review as one indivisible unit risks hiding the information that is most useful for both researchers and managers. For airlines, it is not enough to know that a passenger was dissatisfied. It is more important to understand whether dissatisfaction came from punctuality, comfort, crew behaviour, food, price fairness, ground service, or the interaction between several of these elements.

Aspect-Based Sentiment Analysis addresses this limitation by moving from document-level sentiment to aspect-level interpretation. Instead of asking only whether a review is positive or negative, ABSA asks what is being evaluated and what sentiment is expressed toward that specific aspect. This distinction is central to the present thesis. Airline reviews are not treated simply as texts to be classified, but as structured narratives of service experiences. Each review can contain several service dimensions, each dimension can carry a different sentiment, and each sentiment can be influenced by the wider context in which the journey occurred.

The motivation for this research comes from the gap between the richness of passenger reviews and the limitations of simple analytical approaches. Online reviews contain detailed evidence about service quality, but their unstructured nature makes them difficult to analyse at scale. Structured ratings

provide useful numerical indicators, but they cannot fully explain the reasons behind satisfaction or dissatisfaction. A passenger may give a low rating, but the number alone does not reveal whether the problem was a delay, poor staff behaviour, weak value for money, discomfort, or a combination of these factors. The challenge is therefore to connect textual evidence, structured ratings, and contextual information in a way that preserves the complexity of the passenger experience.

This thesis responds to that challenge by developing a domain-aware ABSA framework for airline passenger reviews. The analysis is based on a large corpus of 144,934 reviews covering 578 airlines and flights from April 2012 to March 2026. The dataset combines written review narratives, overall ratings, service sub-ratings, airline information, verification status, cabin class, traveller type, route, date flown, and other contextual variables when available. This structure makes it possible to analyse passenger satisfaction not as a single outcome, but as a layered judgement formed through language, ratings, service dimensions, and travel context.

The framework is domain-aware because the aspect schema is grounded in the specific structure of the airline experience. The textual analysis focuses on five core service dimensions: Punctuality, Cabin Comfort, Cabin Crew Service, Food & Beverages, and Value For Money. These aspects reflect central points of passenger evaluation: operational reliability, physical comfort, human service, onboard product quality, and perceived fairness between price and service received. This domain-specific design allows the analysis to move beyond generic polarity classification and toward a more meaningful interpretation of airline satisfaction.

The thesis is not limited to extracting aspects and classifying sentiment. It also examines how aspect-level evidence relates to overall satisfaction and

how satisfaction changes across different passenger and operational contexts. The pipeline therefore combines aspect extraction, aspect sentiment classification, rating–aspect alignment, predictive modelling, segment analysis, weather attribution, seasonality, and causal event analysis. This broader design reflects the idea that passenger satisfaction is shaped not only by what the airline provides, but also by when, where, and under which circumstances the journey takes place.

A central methodological objective of the thesis is to compare different levels of modelling complexity. The analysis includes a lexicon baseline, a TF-IDF model with Logistic Regression, and a fine-tuned DistilBERT model. This comparison is important because transformer models are often expected to outperform classical methods in language tasks. However, in applied settings, performance depends not only on model capacity but also on the quality of supervision. In this thesis, aspect sentiment labels are derived from structured ratings rather than from manually annotated gold-standard labels. The comparison between TF-IDF and DistilBERT therefore tests whether additional model complexity is actually useful under noisy, rating-derived silver-label supervision.

The empirical contribution of the thesis is also managerial. Airline managers do not only need general measures of satisfaction; they need to know which parts of the journey create the strongest dissatisfaction, which dimensions are most closely connected to the overall rating, and which passenger segments or travel conditions make service failures more damaging. By connecting textual sentiment with structured ratings and contextual variables, the analysis aims to transform passenger reviews into evidence that can support more targeted service improvement.

The thesis also gives particular attention to contextual factors that are

often outside the scope of standard ABSA research. Weather, seasonality, and major aviation events can all influence how passengers evaluate airline service. A delay caused by severe weather may be interpreted differently from a delay perceived as poor airline management. Peak-season travel may intensify dissatisfaction because airports are more congested, prices are higher, and expectations are emotionally charged. Major disruptions such as pandemics, strikes, safety incidents, or technological outages can affect passenger sentiment beyond the level of individual airlines. For this reason, the thesis extends ABSA beyond classification and uses it as a framework for studying how passengers interpret service quality under changing conditions.

The research is guided by three questions:

- RQ1** How does domain-aware ABSA perform compared to a lexicon baseline, and does fine-tuned DistilBERT improve over TF-IDF on domain-specific airline text?
- RQ2** Which airline service aspects most strongly influence overall customer satisfaction, and how do sentiment distributions vary across aspects, carrier types, cabin classes, and traveller segments?
- RQ3** To what extent do contextual factors—weather, seasonal demand, and global disruption events—modulate the relationship between aspect-level sentiment and overall satisfaction?

The first research question focuses on model performance and methodological suitability. It asks whether a domain-aware supervised approach improves over a simpler lexicon baseline, and whether a transformer model provides additional value compared with a classical TF-IDF classifier. The second research question focuses on the structure of passenger satisfaction.

It asks which aspects matter most and how satisfaction changes across different types of passengers and services. The third research question focuses on context. It asks whether passenger satisfaction changes when the journey is affected by weather, seasonal demand, or major disruption events.

The thesis is organised as follows. Chapter 2 motivates the study of airline passenger satisfaction through online reviews and explains why passenger evaluations are multidimensional. Chapter 3 reviews the development of Aspect-Based Sentiment Analysis, from lexicon-based and classical machine-learning approaches to transformer-based models, and positions the thesis within domain-aware ABSA research. Chapter 4 describes the data source, corpus construction, dataset structure, and exploratory characteristics of the review corpus. Chapter 5 presents the methodological pipeline, including aspect extraction, aspect sentiment classification, predictive modelling, contextual modules, weather attribution, seasonality, and causal event analysis. Chapter 6 reports the empirical results of the pipeline. Chapter 7 answers the three research questions and synthesises the findings across modules. Finally, Chapter 8 concludes the thesis by discussing the main contributions, limitations, future research directions, and managerial implications.

Overall, the purpose of this thesis is to show that airline passenger reviews can be analysed as more than isolated opinions or simple satisfaction scores. When interpreted through a domain-aware ABSA framework, they become a rich source of evidence about how passengers experience, evaluate, and explain airline service quality. The central argument is that satisfaction is not monolithic. It is built across aspects, shaped by expectations, affected by operational reliability, mediated by perceived value, and reinterpreted through context. A framework capable of capturing these layers can provide a deeper understanding of passenger satisfaction than document-level

sentiment analysis or overall ratings alone.

Chapter 2

Understanding Passenger Satisfaction Through Airline Reviews

The airline industry represents one of the most operationally complex and customer-sensitive service sectors in the modern economy. Unlike many industries in which the customer interacts with a single product or a limited number of service points, air travel consists of a long chain of interconnected experiences, each capable of shaping the passenger's perception of the journey as a whole. From the moment a traveller books a ticket to the moment they collect their luggage at the destination airport, satisfaction is continuously influenced by dozens of interactions involving technology, infrastructure, personnel, operational efficiency, comfort, communication, and expectations.

For airlines, customer satisfaction is no longer simply a matter of reputation. It has become a strategic and economic necessity (Parasuraman et al., 1988; Chen and Chang, 2005; Clemes et al., 2008). In a market characterised by intense competition, low switching costs, and growing digital transparency, passengers can easily compare airlines, publicly share their experiences, and influence the decisions of thousands of potential customers through online platforms. A single negative experience may not only result

in the loss of one customer, but may also contribute to reputational damage capable of influencing broader consumer perception. Conversely, positive experiences can strengthen loyalty, increase repurchase intention, and justify premium pricing strategies.

This growing importance of customer experience has fundamentally transformed the role of passenger feedback. Historically, airlines relied heavily on operational indicators such as on-time performance, baggage handling statistics, or internal service evaluations to assess performance quality. While these indicators remain essential, they do not necessarily capture how passengers emotionally experience a journey. A flight may depart only slightly late according to operational standards while still generating strong dissatisfaction among passengers due to stress, uncertainty, or poor communication. Similarly, an airline may deliver technically acceptable service while failing to meet customer expectations relative to price, cabin class, or competing carriers.

This distinction between operational performance and perceived experience is central to understanding why airline reviews are such an important source of information. Passenger reviews provide direct access to the customer perspective in a way that traditional numerical indicators cannot. They reveal not only whether customers were satisfied, but also what specifically influenced their judgement, what frustrations emerged during the journey, and which aspects of the experience passengers considered most important.

The value of reviews lies precisely in their unstructured nature. Unlike surveys with predefined questions and limited response options, reviews allow passengers to freely describe their experiences using their own language, priorities, and emotional framing. In doing so, they often expose dimensions of

dissatisfaction that structured metrics fail to capture. A traveller describing an exhausting delay, an indifferent crew interaction, or the feeling of having overpaid for a disappointing service communicates far more than a numerical score alone ever could. These narratives therefore represent an exceptionally rich source of behavioural and managerial insight.

However, the very richness that makes reviews valuable also makes them difficult to analyse systematically. Passenger experiences are inherently multidimensional, meaning that a single review often contains several distinct evaluations referring to different aspects of the journey. A customer may simultaneously appreciate the professionalism of the cabin crew, criticise the quality of onboard food, complain about a missed connection, and still consider the ticket reasonably priced. Human evaluations are rarely binary, and airline experiences in particular are shaped by a combination of operational, emotional, and contextual factors that interact continuously throughout the journey.

This complexity creates an important limitation for traditional sentiment analysis approaches based on overall positive or negative classification. Treating an entire review as simply “positive” or “negative” inevitably removes the nuance that makes customer feedback useful for decision-making. For airlines, knowing that passengers are dissatisfied is not sufficient. What matters operationally is understanding why they are dissatisfied, which dimensions contribute most strongly to frustration, and which problems generate the greatest impact on overall satisfaction.

This research is motivated precisely by this problem. The objective is not merely to classify sentiment within passenger reviews, but to understand how different dimensions of the airline experience contribute to customer satisfaction and how those dimensions interact with contextual factors such

as delays, weather conditions, seasonality, traveller expectations, and global disruption events. The study therefore approaches reviews not simply as textual data, but as behavioural evidence capable of revealing deeper patterns within passenger perception.

A central motivation of this research is that passenger satisfaction is not formed at a single moment. It develops progressively across the journey, beginning before the passenger reaches the airport and continuing until the trip is completed. The evaluation of an airline is therefore built through a sequence of expectations, interactions, disruptions, and comparisons. For this reason, the dimensions considered in this study were not chosen as isolated variables, but as elements of a wider service experience that reflects how passengers actually perceive air travel.

The first layer of this experience is expectation. Before boarding, the passenger has already made an economic decision: they have selected a fare, a cabin class, a carrier, and sometimes a specific service bundle. From that moment, the journey is evaluated against an implicit question: was this experience worth what I paid? This is why Value for Money is central to the analysis. It does not refer only to price in a narrow sense, but to the perceived balance between cost and quality. A passenger may accept limited service when paying a very low fare, while the same service may be considered unacceptable when associated with a premium price. In this sense, value is not an objective attribute of the flight, but a judgement formed through comparison between expectation and delivery. For airlines, this dimension is particularly important because it connects customer satisfaction directly to pricing strategy, product positioning, and perceived fairness.

Once the journey begins, operational reliability becomes one of the most

decisive elements of the passenger experience. Air travel is strongly time-sensitive: passengers travel to attend meetings, reach holiday destinations, make connections, return home, or participate in important personal events. A delay or cancellation can therefore affect far more than the flight itself. It can create stress, financial loss, missed opportunities, and a sense of helplessness. This is why punctuality occupies a distinctive position within the research. Unlike other service dimensions, it often works as a non-compensable factor. A passenger may tolerate mediocre catering or limited entertainment, but a serious delay can dominate the entire evaluation of the journey. In practical terms, this means that punctuality is not merely one service attribute among others; it is one of the foundations on which trust in the airline is built.

The passenger's evaluation then continues through the physical experience of travel. The cabin is not simply the place where transportation occurs; it is the environment in which passengers spend the most visible and memorable part of the journey. Seat comfort, legroom, cleanliness, cabin temperature, noise, storage space, and the general feeling of physical ease all contribute to how passengers interpret the quality of the flight. These elements become particularly important on medium- and long-haul journeys, where discomfort accumulates over time and can strongly affect the overall perception of the airline. Cabin comfort therefore represents a direct expression of the airline's product quality and of the trade-off between operational efficiency and passenger wellbeing.

At the same time, air travel remains a deeply human service experience. Even in an industry increasingly shaped by automation, digital check-in, self-service kiosks, and algorithmic pricing, passengers still remember how they were treated by people. Cabin crew and ground staff often become the

emotional interface between the customer and the airline, especially when something goes wrong. A helpful, calm, and empathetic crew member can reduce frustration during delays, manage uncertainty, and transform an otherwise negative experience into a more acceptable one. Conversely, a rude or indifferent interaction can intensify dissatisfaction even when the operational problem itself is relatively minor. The human component of service is therefore essential because it influences not only what passengers experience, but how they emotionally interpret that experience.

Onboard services such as food and beverages also contribute to this perception, although in a different way. Catering may not always be the primary determinant of satisfaction, especially on short-haul or low-cost flights, but it often functions as a symbolic indicator of care, quality, and attention to detail. On longer journeys and premium services, food becomes part of the expected service promise. When the catering is poor, passengers may interpret it not simply as a disappointing meal, but as evidence that the airline has reduced quality or failed to respect the expectations associated with the ticket. For this reason, food and beverages are relevant not only as a material service component, but also as part of the broader judgement passengers form about whether the airline values their comfort.

These service dimensions do not operate independently. They interact continuously in the passenger's mind. A comfortable seat may soften the perception of a long flight, but it may not compensate for a missed connection. A friendly crew may reduce frustration, but may not fully repair the damage caused by poor communication during disruption. A low ticket price may make passengers more tolerant of limited service, while a premium fare raises the threshold of what is considered acceptable. This interaction is precisely why an aspect-based approach is necessary. Overall ratings can

indicate whether a passenger was satisfied or dissatisfied, but they cannot explain which part of the journey created that judgement.

The same logic applies to the contextual factors included in the research. Passenger satisfaction is not shaped only by the airline’s direct service delivery, but also by the circumstances surrounding the journey. Weather, for example, introduces an important distinction between outcome and responsibility. A passenger may suffer a delay, but their reaction can change depending on whether they perceive the cause as controllable or uncontrollable. Severe turbulence, storms, fog, or de-icing procedures may be accepted as external constraints, while poor communication or inadequate disruption management may still be blamed on the airline. Analysing weather-related language therefore helps reveal how passengers assign responsibility and how attribution influences satisfaction.

Seasonality adds another layer of complexity. Airlines do not operate under constant conditions throughout the year. During peak travel periods, demand increases, airports become more congested, aircraft utilisation intensifies, and operational slack decreases. These conditions can make service failures more likely and recovery more difficult. For passengers, peak-season travel is often associated with holidays, family trips, and higher emotional investment, which may further amplify dissatisfaction when disruptions occur. Studying seasonality therefore allows the research to examine whether satisfaction declines under demand pressure and whether specific passenger segments are more exposed to this deterioration.

Differences between traveller types and cabin classes are equally important because passengers do not evaluate service from the same position. A business traveller may prioritise punctuality, efficiency, and reliability, while

a leisure traveller may place greater emphasis on comfort, price, or the emotional quality of the journey. Families may experience boarding, seating, delays, and service interactions differently from solo passengers. Similarly, Economy passengers may evaluate the experience through affordability, while premium passengers judge the airline against a higher service promise. These differences mean that average satisfaction alone can hide important inequalities in how service quality is perceived across customer groups.

Finally, the inclusion of major aviation events reflects the fact that the airline industry is highly exposed to external shocks. Pandemics, strikes, airspace closures, safety incidents, and technological disruptions can alter passenger sentiment beyond the level of individual airlines. These events affect expectations, trust, operational stability, and the emotional climate in which passengers evaluate their journeys. By considering such disruptions, the research moves beyond ordinary service evaluation and examines how broader industry conditions influence customer perception over time.

In this sense, the factors examined in this study are connected by a common logic: each represents a point at which the passenger’s experience can be shaped, damaged, repaired, or reinterpreted. Together, they allow the analysis to move from a simple question—whether passengers are satisfied—to a more meaningful managerial question: what exactly creates satisfaction or dissatisfaction, under which conditions, and for which passengers? This is the practical value of the research. By identifying the dimensions that matter most and the contexts in which they become most critical, airlines can move from general customer feedback to targeted improvement strategies.

Another important motivation behind this research is the recognition that customer reviews contain information that extends beyond individual complaints. When analysed collectively and at scale, reviews reveal structural

patterns that may remain invisible through manual inspection alone. A single complaint about poor connectivity may appear anecdotal, but thousands of similar complaints across airlines and years suggest a persistent industry-wide issue. Likewise, repeated references to delays during peak seasons may indicate systemic operational stress rather than isolated events. The ability to detect these patterns transforms reviews from isolated opinions into strategic intelligence.

This research therefore seeks to contribute not only to academic literature, but also to practical managerial understanding. Airlines continuously invest substantial resources into improving customer experience, redesigning cabin products, introducing digital services, and optimising operations. However, investments are most effective when guided by evidence identifying which aspects genuinely influence satisfaction and which problems generate the strongest dissatisfaction. Understanding whether passengers care more about pricing fairness than onboard entertainment, or whether delays outweigh improvements in service quality, has direct implications for resource allocation and strategic priorities.

From a broader perspective, this study also reflects the increasing importance of data-driven decision-making within service industries. Modern organisations generate enormous quantities of customer feedback, yet much of this information remains underutilised because of the difficulty involved in analysing unstructured text systematically. Advances in Natural Language Processing create new opportunities to transform large-scale textual feedback into measurable and actionable insights. Within the airline sector, where customer experience directly influences competitive positioning, the ability to extract meaningful information from passenger reviews represents an increasingly valuable capability.

Ultimately, airline reviews are important not simply because they contain opinions, but because they capture the relationship between operational reality and human perception. They reveal how passengers interpret service quality, how expectations shape evaluations, and how external conditions influence customer judgement. By analysing these reviews through a detailed and context-aware framework, this research aims to provide a deeper understanding of passenger satisfaction and to demonstrate how textual feedback can support airlines in identifying weaknesses, improving service quality, and making more informed strategic decisions.

Chapter 3

Related Work and State of the Art

The literature relevant to this thesis spans two research traditions that are highly complementary but have often developed separately. The first is Aspect-Based Sentiment Analysis, a subfield of Natural Language Processing concerned with identifying the targets and polarities of opinions in text. The second is the empirical study of airline passenger satisfaction, which has produced rich managerial and service-quality frameworks for understanding how travellers evaluate airlines across multiple dimensions. Bringing these traditions together is useful because airline reviews are not only textual data. They are also narratives of service experience, in which passengers describe what happened, what they expected, who they blamed, and which part of the journey shaped their final judgement.

This chapter therefore does not review the literature as a sequence of isolated works. Instead, it uses the literature to frame the problem addressed by the thesis. The first section discusses the general state of the art on ABSA, focusing on the movement from document-level sentiment to aspect-level interpretation and on the methodological trade-off between transparency, model capacity, and label quality. The second section turns to airline passenger

satisfaction and online reviews, showing why the aviation domain requires a domain-specific and context-aware interpretation of sentiment. The final section briefly positions the present thesis relative to both traditions, explaining what is new in the proposed framework and why it matters from both a methodological and managerial perspective.

3.1 General State of the Art on Aspect-Based Sentiment Analysis

The central claim of Aspect-Based Sentiment Analysis is that opinions have internal structure, and that this structure is analytically consequential. Early sentiment classifiers (Pang et al., 2002; Turney, 2002) demonstrated that the evaluative tone of a document could be estimated computationally, but a global polarity score becomes insufficient when the objective is to understand why a service succeeds or fails along specific dimensions. A passenger review that praises the crew while condemning punctuality is not simply a moderately positive or moderately negative review. It is structured evidence about two distinct components of the service experience, and any single-label representation discards that distinction.

ABSA emerged precisely to address this limitation. Instead of assigning sentiment to an entire document, it asks what is being evaluated and what sentiment is expressed toward that target. The SemEval shared tasks (Pontiki et al., 2014, 2016) formalised this idea into a benchmark framework by distinguishing between aspect extraction, aspect category detection, and aspect sentiment classification. Their contribution was fundamental because they gave the field common tasks, datasets, and evaluation procedures. At the same time, they also helped shape the boundaries of the field: much

ABSA research has been developed around a limited set of benchmark domains, especially restaurants, laptops, and product reviews. These domains are useful for model comparison, but they do not fully capture the operational and narrative complexity of air travel.

Within ABSA, the choice of modelling approach involves a tension that runs through the literature: transparency versus representational power. Lexicon-based approaches (Taboada et al., 2011) make predictions auditable because each decision can often be traced back to a word or rule. This is valuable for interpretation, especially in managerial settings where users may want to understand why a model classified a sentence in a certain way. Their limitation is that sentiment is rarely only a property of individual words. In airline reviews, a passenger may express severe dissatisfaction through a factual description such as waiting on the tarmac for three hours with no information. No explicitly negative adjective is required for the experience to be negative. For this reason, lexicon methods are useful as interpretable baselines, but they are weak when sentiment is implicit, event-based, or context-dependent.

Classical supervised methods address part of this limitation by learning associations between textual features and sentiment labels from data. Bag-of-words representations, n-gram features, and TF-IDF vectors allow classifiers to identify patterns that predefined dictionaries would miss (Kiritchenko et al., 2014). Among these approaches, TF-IDF combined with Logistic Regression remains especially relevant. It is computationally efficient, relatively interpretable, and often competitive in applied corpora. Its usefulness is not only historical. When labels are noisy or indirectly derived, as in the case of rating-based silver labels, a simpler supervised model can sometimes be more robust than a larger model that has enough capacity to fit inconsistencies in

the training signal.

Transformer-based models represent the opposite side of the same methodological trade-off. Models such as DistilBERT (Sanh et al., 2019) produce contextual representations in which the meaning of a word depends on the sentence in which it appears. This is highly relevant for sentiment analysis because polarity often depends on context. Sun et al. (2019) showed how BERT-style models can be adapted to aspect sentiment classification by conditioning the model on the target aspect, so that the task is no longer to classify the whole text but to classify sentiment toward a specific service dimension. This is conceptually aligned with the objective of ABSA. Yet greater representational capacity does not automatically imply better empirical performance. If the supervision signal is ambiguous or noisy, a transformer may learn the noise more effectively rather than recover a cleaner sentiment structure. This is one of the reasons why the present thesis compares TF-IDF and DistilBERT not as an old-versus-new contrast, but as a test of how model capacity interacts with silver-label quality.

Aspect extraction introduces a further challenge. Mapping a textual span to an aspect category is not a simple lexical matching problem, because passengers rarely use canonical category names. They write about missed connections rather than “Punctuality”, or about narrow legroom and cramped seats rather than “Cabin Comfort”. Aspect extraction has therefore often been formulated as a sequence-labelling task. Early feature-based opinion mining (Hu and Liu, 2004) already recognised the importance of organising opinions around the features being evaluated, while Jakob and Gurevych (2010) showed how Conditional Random Fields could be used to identify opinion targets in context. Neural architectures later extended this logic by combining representation learning with structured prediction (Lample et al.,

[2016]). This line of work is directly relevant to the present thesis, where aspect extraction is treated through BIO tagging in order to identify the spans through which passengers refer to airline service dimensions.

Aspect sentiment classification is closely related to extraction because polarity becomes meaningful only once it is anchored to a target. A sentence may contain several aspects with different sentiments, and a model must avoid mixing them. Attention-based models ([Wang et al., 2016]) addressed this problem by allowing the model to focus on different parts of the sentence depending on the target aspect. Joint frameworks such as [Li et al. (2019)] pushed this further by modelling target extraction and sentiment prediction together. The present thesis keeps extraction and classification as separate modules for transparency and interpretability, but follows the same underlying principle: the sentiment signal is only useful if it remains connected to the specific service dimension that generated it.

Despite these developments, the ABSA literature has not fully resolved the problem of domain transfer. Surveys such as [Zhang et al. (2022)] and [Nazir et al. (2020)] identify domain shift, implicit aspects, annotation quality, and contextual variation as persistent challenges. These issues are especially visible in the airline domain. A model trained on restaurant or laptop reviews has little reason to interpret a missed connection, a de-icing procedure, a long taxi time, or a fare-service mismatch as sentiment-bearing evidence. The problem is not only that the vocabulary changes. The structure of the experience changes. This motivates treating airline reviews not as a simple application site for generic ABSA tools, but as a domain requiring its own aspect schema, its own labelling logic, and its own interpretation of model performance.

3.2 Online Reviews and Passenger Satisfaction in the Airline Domain

The airline satisfaction literature establishes a point that is directly relevant to this thesis: passenger satisfaction is not a single quantity, but the aggregate outcome of several service dimensions that travellers evaluate separately and then combine into an overall judgement. This is consistent with the broader service-quality tradition, in which satisfaction is understood as the relationship between expectations and perceived performance (Parasuraman et al., 1988). Airline-specific studies similarly show that passengers evaluate punctuality, staff behaviour, comfort, food, and perceived value as recurring dimensions of service quality (Chen and Chang, 2005; Clemes et al., 2008). The implication for text analysis is direct. A model that assigns one label to a whole review discards much of the information that makes passenger feedback useful for decision-making.

Online reviews extend this literature because they provide unsolicited passenger narratives rather than answers to predefined survey questions. They allow travellers to describe the journey in their own words, often combining operational details, emotional reactions, comparisons with expectations, and judgements of responsibility. Studies such as Sezgen et al. (2019) and Lucini et al. (2020) show that airline reviews can reveal systematic patterns in passenger evaluation, while Peng et al. (2020) highlights the importance of understanding which service dimensions matter most to customers. These studies support the idea that airline reviews are not merely anecdotal complaints. When analysed at scale, they become a source of structured evidence about service quality.

At the same time, much of the previous airline-review literature does not fully integrate the level of detail required by ABSA with the contextual

interpretation required by service-management research. Some studies focus on general sentiment or topic discovery, while others identify satisfaction drivers without assigning sentiment to specific service aspects. The gap is therefore not simply the absence of airline data in NLP. It is the lack of an integrated framework able to extract aspect-level evidence from large-scale reviews, relate it to structured ratings, and interpret it through the operational and managerial context of air travel.

One important contextual factor is causal attribution. A delay is not only an operational event; it is also an event for which passengers assign responsibility. [Weiner \(1985\)](#) argues that individuals respond to outcomes partly by attributing causes, especially by judging whether the cause is controllable. In consumer behaviour, [Folkes \(1984\)](#) shows that reactions to service failure depend on perceived responsibility, while [Hess et al. \(2003\)](#) emphasise the importance of recovery strategies and justice perceptions. In airline reviews, this means that a delay attributed to severe weather may generate a different response from a delay attributed to poor airline organisation. Passengers often encode this attribution directly in the language of their reviews, making weather-related vocabulary an analytical signal rather than a purely descriptive detail.

Seasonality introduces a second contextual layer. Airlines operate under changing demand conditions throughout the year. Peak travel periods increase passenger volume, reduce operational slack, and raise expectations at precisely the moment when service delivery becomes more difficult. Tourism research has long treated seasonality as a structural issue ([Baum and Lundtorp, 1999](#); [Lundtorp et al., 1999](#)), and this logic applies directly to aviation. A seasonal decline in satisfaction should therefore not be interpreted only as a fluctuation in passenger mood. It may reflect the pressure that demand

places on the airline service system, especially during periods when delays, congestion, and recovery constraints become more likely.

Perceived value is a third important factor. Passengers do not evaluate service quality in absolute terms. They evaluate it relative to the fare paid, the cabin class chosen, the route, the carrier type, and the implicit promise of the airline’s market positioning. Service-management research emphasises that satisfaction depends on the comparison between expected and perceived quality (Gilbert et al., 2003). In the airline domain, this comparison is captured strongly by Value For Money. A limited service may be acceptable when the fare is low, but unacceptable when the passenger paid a premium price or expected a higher level of care. This makes perceived value not only a commercial variable, but also a fairness judgement.

Using online reviews also requires methodological caution. Passengers who write public reviews are not a random sample of all passengers. Extreme experiences, both positive and negative, are often overrepresented (Hu et al., 2009; Luca and Zervas, 2016). The corpus used in this thesis should therefore be interpreted as a large-scale record of experiences strong enough to motivate public narration, not as a perfectly representative survey of all airline customers. This limitation does not invalidate the analysis, but it defines its interpretation: review data are especially valuable for understanding the language, structure, and intensity of satisfaction and dissatisfaction.

A further methodological issue concerns event analysis. Major aviation events, such as strikes, regulatory changes, safety incidents, or technological disruptions, may influence passenger satisfaction, but their effect cannot be inferred reliably through a simple before-and-after comparison. Satisfaction also changes with seasonality, demand pressure, and broader industry trends. Difference-in-Differences, associated with Card and Krueger (1994), provides

a more disciplined way to compare changes in an affected period with changes in a control period. In the present thesis, this logic is used in Chapter 6 to examine seven aviation events while reducing the risk of confusing event effects with background temporal variation.

Taken together, the airline satisfaction literature and the online-review literature show that passenger reviews are rich but complex evidence. They reveal how travellers describe service quality, but they are shaped by self-selection, expectations, attribution, seasonality, and value judgements. This is why a generic sentiment classifier is not enough. The analysis must identify the service dimension being discussed, assign sentiment to that dimension, connect the signal to structured ratings, and interpret the result in light of the conditions surrounding the journey.

3.3 Positioning and Novelty of the Present Thesis

The two bodies of literature reviewed above have often asked different questions about related phenomena. ABSA research has focused on how to identify opinion targets and polarities, usually in benchmark domains with relatively clean labels and established schemas. Airline satisfaction research has focused on which service dimensions shape passenger evaluations, often through structured surveys or more general review analysis. The present thesis brings these questions together by designing a domain-aware ABSA framework for airline reviews, where the aspect schema, modelling choices, and contextual analyses are developed around the structure of the passenger experience.

The thesis contributes to the literature in three main ways. First, it defines an airline-specific aspect schema covering Punctuality, Cabin Comfort,

Cabin Crew Service, Food & Beverages, and Value For Money, and applies it to a large corpus of 144,934 reviews covering 578 airlines. Second, it compares a lexicon baseline, TF-IDF with Logistic Regression, and fine-tuned DistilBERT under rating-derived silver-label supervision, showing that model complexity must be interpreted in relation to label quality. Third, it extends ABSA beyond extraction and classification by linking aspect-level sentiment to overall satisfaction, predictive modelling, weather attribution, hemisphere-aware seasonality, traveller segmentation, and Difference-in-Differences event analysis.

This positioning makes the thesis more than a technical NLP exercise. ABSA is used as a structured lens for transforming passenger narratives into evidence about service quality, operational reliability, perceived fairness, service recovery, expectation management, and contextual pressure. The novelty of the work lies in this integration: it connects computational text analysis with the managerial problem of understanding not only whether passengers are satisfied, but which aspects matter most, under which conditions dissatisfaction intensifies, and how passengers interpret the responsibility for service failures.

Chapter 4

Data: Collection, Structure, and Exploration

4.1 Data Source and Corpus Construction

The empirical foundation of the analysis is a large corpus of airline passenger reviews collected from airlinequality.com, the public review platform associated with SkyTrax. This source was selected because it provides a type of data that is particularly suitable for studying passenger satisfaction in a multidimensional service industry. Airline reviews are not only numerical judgements. They combine written narratives, overall ratings, service-specific sub-ratings, and contextual information about the journey. This structure makes them especially valuable for Aspect-Based Sentiment Analysis because the text reveals how passengers describe their experiences, while the ratings provide a structured indication of how they evaluated different parts of the service.

The dataset can therefore be understood as a hybrid source of information. The first component is unstructured textual data: the review body written by the passenger. This is where passengers explain what happened, what disappointed them, what exceeded expectations, and which moments

shaped their final judgement. The second component is structured numerical data: the overall rating and the service sub-ratings. These variables make it possible to compare the written narrative with the passenger’s explicit evaluation of the flight. The third component is contextual data, including information such as airline, cabin class, traveller type, route, verification status, and date flown. Taken together, these elements allow the corpus to represent passenger experience not as a single score, but as a combination of language, evaluation, and travel context.

This type of data is important because airline satisfaction is rarely produced by one isolated factor. A passenger may give a low overall score because of a delay, while still describing the cabin crew as professional. Another passenger may report a comfortable seat and good onboard service, but still feel that the price paid was not justified. A third passenger may be satisfied overall, but complain that the onboard WiFi did not work. These situations are precisely why airline reviews are useful for this research: they contain both the complexity of the passenger narrative and the numerical structure needed to study satisfaction systematically.

The reviews were collected through an automated procedure and then cleaned before being used as the final analytical corpus. The raw scrape produced 149,265 rows from 581 airlines. As is common with web-collected data, the raw observations contained repeated entries, incomplete records, and inconsistent fields that had to be standardised. The cleaning process normalised the relevant variables, removed duplicate observations based on airline, date, and title combinations, and excluded records with missing overall ratings. The overall rating is necessary because it provides the global satisfaction measure against which the service-specific evaluations can later be interpreted. After cleaning, the final usable corpus contained 144,934 reviews

covering 578 airlines.

The platform applies a manual moderation process before publishing reviews. This introduces a delay between the moment a passenger submits a review and the moment that review appears online. For this reason, although the data collection was conducted in March 2026, the newest published review in the corpus dates to September 2025. This gap does not represent a weakness of the collection process; rather, it reflects the publication process of the source platform. It is nevertheless important to acknowledge because the corpus is recent, but it does not include the very latest passenger experiences at the exact time of collection.

4.2 Structure and Characteristics of the Dataset

Table 1: Dataset summary statistics.

Property	Value
Total reviews	144,934
Airlines covered	578
Date range (flown)	April 2012 – March 2026
Newest published review	September 2025
Sub-rating fill rate	98.4%
Mean overall rating	4.38 / 10
Verified reviews	68.3%

Table 1 summarises the main characteristics of the final corpus. The first important element is scale. With 144,934 usable reviews, the dataset is large enough to support both textual analysis and statistical exploration. This matters because airline reviews are highly variable: passengers differ by route, airline, cabin class, travel purpose, expectations, and reason for writing. A small dataset could easily overrepresent a narrow group of experiences, while

a corpus of this size makes it possible to observe broader patterns across many carriers and many years.

The second important element is coverage. The corpus includes 578 airlines, which means that the data are not limited to a single company, country, or business model. It contains reviews of large international airlines, regional carriers, low-cost carriers, and smaller operators. This variety is useful because the objective is not to describe one airline in isolation, but to study how passengers evaluate airline service dimensions across a broad aviation context. The dataset therefore provides a wide empirical base for analysing satisfaction patterns that are not tied exclusively to one carrier.

The date range also gives the corpus analytical value. Reviews cover flights from April 2012 to March 2026, allowing the data to capture more than a decade of passenger experiences. This period includes ordinary operating years, the COVID-19 disruption, the post-pandemic recovery, periods of labour unrest, changes in demand, and several major aviation events. The temporal depth of the corpus is therefore important because passenger satisfaction is not static. It changes with industry conditions, operational pressure, and external shocks.

The high sub-rating fill rate is another strength of the dataset. Each review includes an overall rating on a 1–10 scale and up to seven service sub-ratings on a 1–5 scale: Seat Comfort, Cabin Staff Service, Food & Beverages, Ground Service, Inflight Entertainment, Wifi & Connectivity, and Value For Money. These sub-ratings are important because they show how passengers divide the airline experience into different service dimensions. The overall rating indicates the final judgement, but the sub-ratings help explain the internal structure of that judgement.

The verified-review share, equal to 68.3%, also contributes to the credibility of the corpus. Verification does not remove all limitations of online review data, since passengers still self-select into writing, but it strengthens the reliability of a substantial portion of the observations. At the same time, the dataset should not be interpreted as a perfectly representative survey of all airline passengers. It is a corpus of public reviews written by passengers who were motivated to share their experience. This distinction is essential for interpreting the rating distribution later in the chapter.

A few anonymised and condensed examples illustrate the kind of information contained in the records. One Economy Class solo-leisure review gives an overall rating of 1/10, with low scores for Ground Service and Value For Money. The narrative describes confusion at a small airport, a changed departure time, and poor communication, while also noting that the short flight itself had no further major problems. This example shows that dissatisfaction may originate before boarding rather than during the flight. It also shows why the route from text to satisfaction is not always direct: the operational experience around the flight can dominate the passenger’s overall judgement even when the in-flight component is relatively uneventful.

A second example shows the opposite pattern. An Economy Class family-leisure review gives an overall rating of 9/10, with very high scores for Seat Comfort, Cabin Staff Service, Food & Beverages, Ground Service, and Value For Money, but a low score for Wifi & Connectivity. The passenger describes the flight as punctual, the staff as kind, and the onboard service as generous, while the connectivity dimension remains weak. This type of record is important because it shows that even a very positive review can contain a specific negative aspect. A global positive rating would hide this detail, while an aspect-based reading preserves it.

A third example is more mixed. An Economy Class solo-leisure review gives a medium overall rating, with good scores for comfort, crew, food, and value, but weak scores for entertainment and connectivity. The narrative describes the general experience as acceptable while expressing dissatisfaction with the entertainment options on a long-haul journey. This kind of review is useful because it shows that neutral or moderate ratings are not empty middle points. They often reflect a balance between acceptable service dimensions and specific weaknesses. Such cases are important for understanding why aspect-level sentiment is more informative than overall polarity alone.

These examples are not intended to represent the entire corpus, but they make the structure of the data more concrete. Each review can combine several dimensions of the journey, and the same passenger can express different evaluations toward different aspects. This is precisely what makes the dataset appropriate for the research design: it provides both the narrative complexity needed for textual analysis and the structured ratings needed to interpret satisfaction across service dimensions.

4.3 Exploratory Description of the Corpus

The first exploratory step examines the distribution of overall ratings and review length. This is necessary because the structure of the data influences how the corpus should be interpreted. A voluntary review platform does not behave like a random passenger survey. Travellers decide for themselves whether to write a review, and that decision is often influenced by the intensity of the experience. Extremely negative and extremely positive journeys are more likely to be documented, while routine experiences often remain invisible.

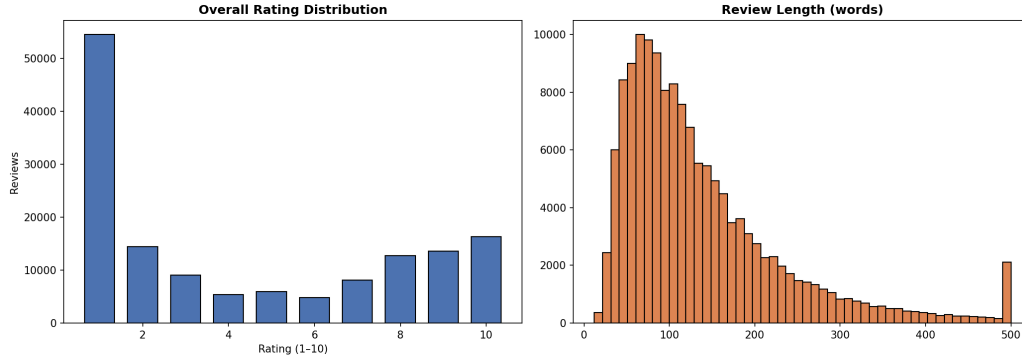


Figure 1: Overall rating distribution and review length across the corpus. The strongly bimodal shape reflects the self-selection dynamics of voluntary review platforms.

Figure 1 shows this dynamic clearly. The left panel displays the overall rating distribution, and the right panel shows review length in words. The rating distribution is strongly bimodal. Rating 1 is by far the most frequent category, with approximately 54,000 reviews. At the same time, high ratings are also well represented, especially ratings 8, 9, and 10. The middle of the scale is much thinner, with ratings 4, 5, and 6 appearing less frequently.

This pattern is important because it reveals that the corpus is shaped by self-selection. Passengers are more likely to write when the experience feels exceptional, either negatively or positively. A severe delay, a missed connection, a lost bag, or a rude interaction can create a strong motivation to write publicly. A very positive experience can have the same effect when service exceeds expectations. By contrast, a normal flight that is neither especially good nor especially bad may not produce enough motivation for the passenger to leave a review. This is consistent with the review-platform dynamics discussed by Hu et al. (2009), where online ratings often overrepresent extreme experiences.

The implication is that the mean overall rating of 4.38 out of 10 should

be interpreted carefully. It does not necessarily represent the average satisfaction of all passengers who travelled during the period. It represents the average rating of passengers who chose to publish a review on this platform. The difference matters because the corpus is best understood as a record of memorable or noteworthy experiences rather than as a probability sample of all airline customers.

The right panel of Figure 1 adds an equally important observation. Most reviews contain enough words to provide meaningful textual evidence. The distribution is right-skewed, with many reviews concentrated in the shorter-to-medium range and a long tail of much longer narratives. This is useful for the analysis because ABSA requires more than a numerical rating; it requires textual material from which aspects and opinions can be extracted. The presence of longer reviews is particularly valuable because passengers often explain complex experiences in detail when disruption, disappointment, or strong satisfaction is involved.

After the overall rating distribution, the next step is to examine the seven service sub-ratings. These variables provide an initial view of which dimensions of the airline experience are evaluated more favourably and which appear more problematic.

Table 2: Mean sub-ratings across all reviews (scale 1–5).

Dimension	Mean	Rank
Cabin Staff Service	3.01	1 (highest)
Seat Comfort	2.74	2
Food & Beverages	2.73	3
Inflight Entertainment	2.64	4
Value For Money	2.64	5
Ground Service	2.39	6
Wifi & Connectivity	1.98	7 (lowest)

Table 2 shows that satisfaction is not evenly distributed across service dimensions. Cabin Staff Service has the highest mean score, at 3.01 out of 5. This does not mean that passengers are fully satisfied with cabin crew, but it suggests that the human service component is evaluated more favourably than other parts of the experience. This is meaningful because cabin crew are often the most visible representatives of the airline during the flight. Even when operational problems occur, professional and empathetic staff can soften the passenger’s perception of the journey.

Seat Comfort and Food & Beverages follow with mean scores of 2.74 and 2.73. These values suggest moderate dissatisfaction rather than strong approval. They indicate that the physical and onboard product remains important, but these dimensions are not the lowest-performing areas in the corpus. Inflight Entertainment and Value For Money both have a mean of 2.64. The position of Value For Money is particularly relevant because it reflects the passenger’s judgement of whether the experience was worth the price paid. Even before deeper modelling, its relatively low score indicates that many passengers question the balance between cost and delivered service.

Ground Service has a mean score of 2.39, making it the second-lowest dimension. This is important because the airline experience begins before the aircraft takes off. Check-in, boarding, gate communication, baggage handling, transfers, and disruption management all shape the passenger’s perception of the airline. A low Ground Service score suggests that dissatisfaction often emerges around the organisational and communication processes surrounding the flight, not only inside the cabin.

The lowest score belongs to Wifi & Connectivity, with a mean of 1.98. It is the only dimension below 2.0. This result is striking because it identifies digital connectivity as the weakest service area in the corpus. The low

value may reflect unavailable WiFi, unreliable connection, slow speed, high price, or a mismatch between passenger expectations and the service actually provided. In modern travel, connectivity is increasingly perceived as part of the expected service environment, especially for business travellers and long-haul passengers. The low score therefore suggests a persistent gap between passenger expectations and airline delivery.

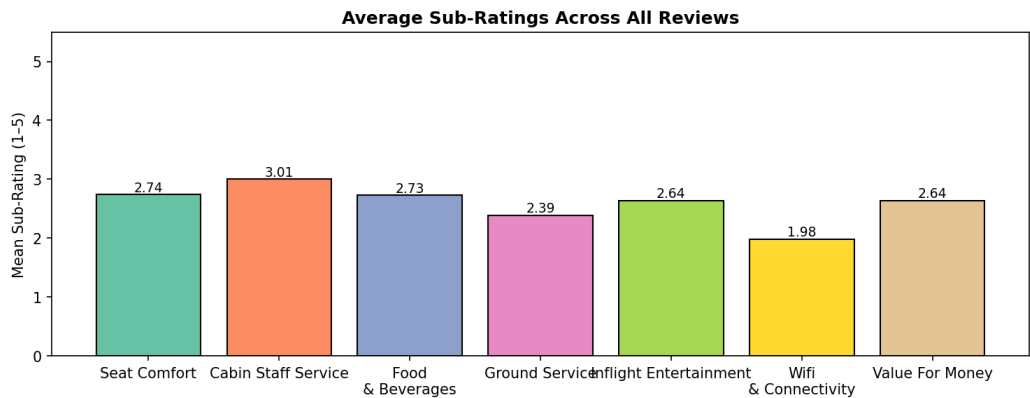


Figure 2: Average passenger sub-ratings across the seven service dimensions.

Figure 2 makes the ranking in Table 2 visually immediate. Most dimensions cluster between roughly 2.6 and 3.0, while Ground Service and Wifi & Connectivity fall below the rest. The figure therefore shows that dissatisfaction is not randomly distributed. Certain service areas appear systematically weaker than others.

The visual contrast between Cabin Staff Service and Wifi & Connectivity is especially informative. Cabin Staff Service, the most human and interpersonal dimension, receives the highest average score. Wifi & Connectivity, a technological and infrastructural dimension, receives the lowest. This contrast suggests that passengers may evaluate direct human interaction more favourably than the systems and services surrounding the journey. From a managerial perspective, this distinction matters because different weaknesses

require different types of intervention. Improving staff service involves training, culture, and service recovery. Improving connectivity requires technological reliability, investment, and clearer product expectations.

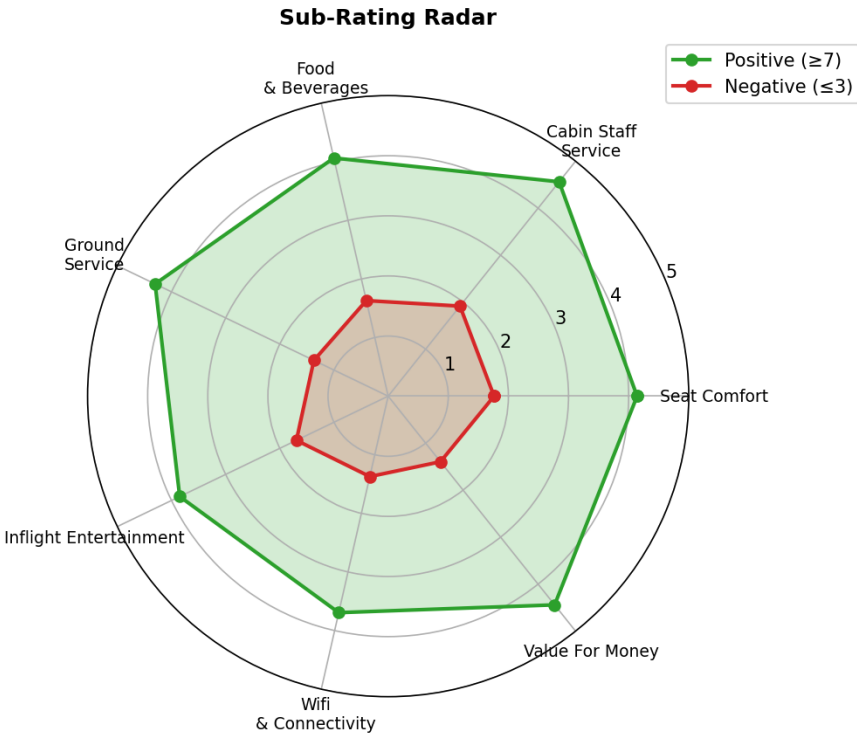


Figure 3: Sub-rating radar chart highlighting average passenger satisfaction across service dimensions.

Figure 3 provides a complementary view of the same pattern. The radar chart represents the seven service dimensions as a satisfaction profile. If passenger evaluations were balanced across all dimensions, the shape would be more regular. Instead, the profile is uneven. Cabin Staff Service extends further outward, while Wifi & Connectivity pulls the shape inward. This unevenness visually reinforces the central assumption of the research design: airline satisfaction is multidimensional.

The radar chart is useful because it shows why an overall rating alone

is insufficient. A single score cannot reveal whether dissatisfaction is driven by ground processes, connectivity, seat comfort, perceived value, or onboard service. The sub-rating profile shows that passengers differentiate between these dimensions, and that the airline experience cannot be treated as one homogeneous object. This is exactly the type of structure that an aspect-based approach is designed to analyse at the textual level.

Taken together, the exploratory analysis shows that the corpus is both rich and complex. It is rich because it contains a large number of reviews, broad airline coverage, structured ratings, and detailed passenger narratives. It is complex because it comes from a voluntary review platform, where extreme experiences are more likely to appear than routine ones. The data therefore need to be interpreted as expressed passenger satisfaction rather than as a perfectly representative measure of all passenger satisfaction.

This chapter establishes the empirical foundation for the analysis that follows. The dataset is suitable because it combines the three elements required for a domain-aware study of airline reviews: natural-language descriptions of passenger experience, structured service evaluations, and contextual information about the journey. The first descriptive patterns already show that passenger satisfaction is uneven across service dimensions, that reviews are shaped by self-selection, and that specific weaknesses such as Ground Service and Wifi & Connectivity emerge clearly before any modelling is performed. This provides the basis for the methodological framework developed in the next chapter.

Chapter 5

Methodology

The methodological framework is designed as an end-to-end pipeline that transforms raw passenger reviews into structured evidence about airline service quality. The starting point is a large collection of written reviews and numerical ratings; the final output is a set of aspect-level, contextual, predictive, and managerial insights. Rather than treating passenger feedback as a single sentiment score, the pipeline follows the logic of the passenger journey. It first prepares and explores the review corpus, then identifies the service dimensions discussed in the text, assigns sentiment to those dimensions, connects them to passenger ratings, and finally studies how external conditions such as delay, route type, aircraft type, weather, seasonality, traveller profile, and major aviation events influence satisfaction.

The implementation is organised as a modular Python workflow. The main script coordinates the execution of the pipeline, while the configuration file stores shared paths, constants, lexicons, aspect definitions, and reusable settings. Each analytical stage is implemented as a separate module, so that the workflow remains transparent, reproducible, and easy to inspect.¹ This

¹The full code repository is available at: https://github.com/BalerionIT/airline_absa.

modular design is important because the analysis combines several methodological layers: text processing, rule-based aspect extraction, supervised sentiment classification, statistical modelling, contextual segmentation, causal inference, and transformer-based benchmarking.

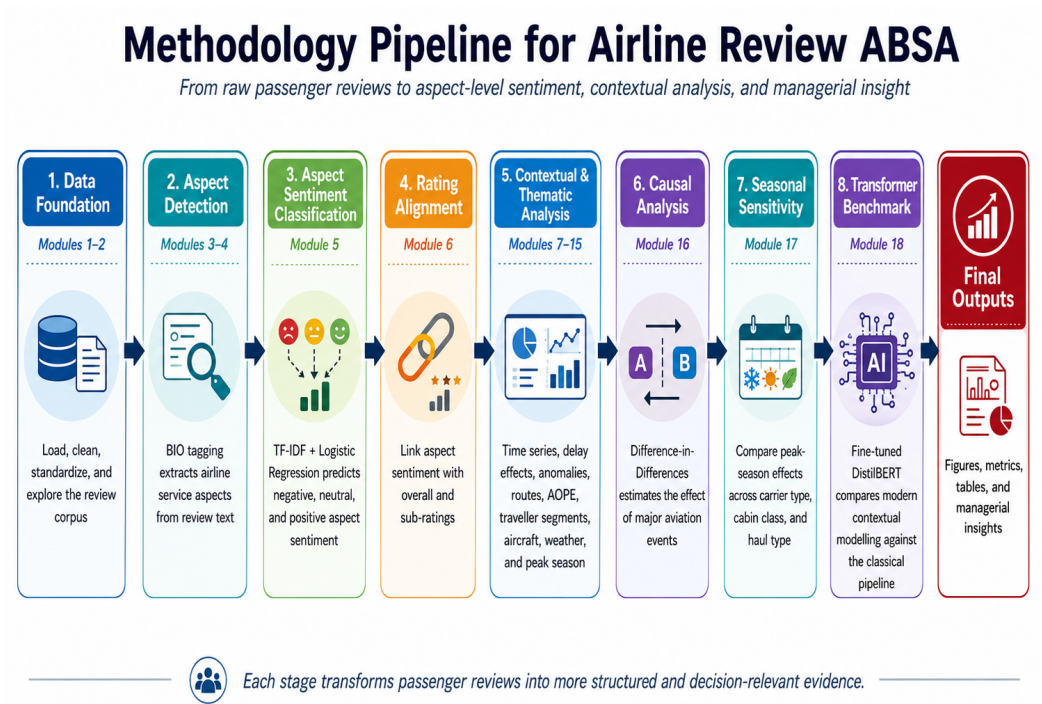


Figure 4: Methodology pipeline for Airline Review ABSA, from raw passenger reviews to aspect-level sentiment, contextual analysis, and managerial insight.

Figure 4 provides a compact overview of the full workflow. The pipeline is deliberately linear: each stage receives the output of the previous one and transforms it into a more structured and decision-relevant form of evidence. The first modules establish the data foundation, the central modules perform the core Aspect-Based Sentiment Analysis, and the later modules extend the analysis toward business questions such as disruption impact, seasonal pressure, passenger segmentation, attribution, and model robustness.

In this sense, the methodology moves from passenger voices to managerial interpretation.

The first module loads the review corpus and prepares it for analysis. At this stage, the data are treated as heterogeneous records that need to be made consistent before any modelling can take place. Column names are standardised, ratings are converted into numerical format, date fields are parsed, missing variables are handled, and observations that cannot support the later analysis are removed. This step is essential because the entire workflow depends on the correct alignment between three elements: the written review, the overall rating, and the service-specific sub-ratings. If these elements are not aligned, the relationship between passenger language and satisfaction would become unreliable.

The same module creates derived variables that are reused throughout the analysis. Review length is computed, recommendation information is normalised when available, date-based variables are created, and rating-based sentiment indicators are generated. A simple lexicon score is also produced by counting positive and negative terms in the review text. This lexicon score is not intended to be the final sentiment model. Its role is to provide a transparent reference point and to show how much can be learned from a simple dictionary-based reading of the text before more domain-aware models are introduced. In managerial terms, this first module turns a raw review file into a usable analytical base.

The second module performs exploratory data analysis. Before extracting aspects or training classifiers, the corpus needs to be understood as a behavioural object. The module examines the distribution of overall ratings, the length of reviews, and the average values of the service sub-ratings. This

stage is not only descriptive. It helps establish how passengers use the platform, how much textual information is available, and whether the ratings suggest a multidimensional satisfaction structure. It also provides the first indication that online reviews should be interpreted carefully, because voluntary review platforms often reflect experiences that passengers felt strongly enough to report publicly.

The exploratory module produces the first figures and tables of the project. These outputs help the reader understand the shape of the dataset before the more advanced analysis begins. The rating distribution gives an initial view of review behaviour, the review-length distribution confirms whether the text is rich enough for aspect extraction, and the sub-rating averages show whether passengers evaluate all service dimensions similarly or whether some dimensions appear systematically stronger or weaker. This stage therefore prepares the ground for the rest of the methodology: it shows why the review corpus is suitable for aspect-level analysis and why a single overall rating cannot fully explain passenger satisfaction.

The third and fourth modules perform aspect extraction, which is the first core ABSA operation. The goal is to identify where passengers mention specific airline service dimensions inside the review text. The aspect schema contains five domain-relevant categories: Punctuality, Cabin Comfort, Cabin Crew Service, Food & Beverages, and Value For Money. These categories were selected because they correspond to major components of the passenger journey. Punctuality captures delays, cancellations, missed connections, and operational reliability. Cabin Comfort captures seats, legroom, cabin environment, and physical space. Cabin Crew Service captures staff behaviour, helpfulness, friendliness, and rudeness. Food & Beverages captures meals, drinks, catering, and onboard provisions. Value For Money captures

price perception, fare fairness, and the relationship between what passengers paid and what they received.

The extraction method is based on BIO sequence tagging. In this approach, the review is treated as a sequence of tokens, and each token is assigned a role: beginning of an aspect expression, continuation of an aspect expression, or outside any aspect expression. This is appropriate because many airline aspects are expressed through phrases rather than isolated words. Expressions such as “missed connection”, “flight attendant”, “seat comfort”, and “value for money” need to be captured as complete units rather than fragmented into separate words. The procedure applies longest-match priority, so longer and more informative expressions are detected before shorter overlapping terms. This keeps the extraction transparent and reproducible while still capturing the phrasing passengers naturally use.

Once the aspect expressions are detected, they are converted into aspect spans and linked to their local textual context. This context is important because sentiment is usually local. A review may criticise punctuality in one sentence, praise the crew in another, and complain about price in a third. If the entire review were classified as positive or negative, these different signals would be mixed together. By extracting aspect-specific contexts, the pipeline allows one review to contribute several separate service evaluations. This is precisely the structure needed for airline reviews, where satisfaction is often mixed rather than uniformly positive or negative.

The fifth module performs Aspect Sentiment Classification. After the pipeline has identified which service dimension is being discussed, it must determine whether the sentiment toward that aspect is negative, neutral, or positive. The main classical model combines TF-IDF text representation with Logistic Regression. TF-IDF transforms the local review context

into weighted textual features, giving more importance to words and phrases that are informative within the corpus. The representation includes both unigrams and bigrams, allowing the classifier to learn individual words as well as short expressions such as “on time”, “no compensation”, “rude staff”, or “missed connection”.

Logistic Regression is used because it is efficient, interpretable, and strong enough to serve as a serious classical benchmark. The model includes regularisation to reduce overfitting and is trained separately for each aspect. This design choice matters because the language of sentiment changes across service dimensions. Negative Punctuality often appears through operational vocabulary related to delays and cancellations, while negative Value For Money may appear through price, cost, fairness, or disappointment with the exchange between fare and service. Training one classifier per aspect allows each model to learn the vocabulary and decision boundary most relevant to that specific service dimension.

Because manually annotated aspect-level sentiment labels are not available at this scale, the classifier is trained using silver labels derived from structured ratings. When a direct sub-rating exists for an aspect, that rating is converted into a sentiment class. Low scores are treated as negative, intermediate scores as neutral, and higher scores as positive. When no direct sub-rating exists, the overall rating is used as a fallback. This strategy makes large-scale supervised learning possible without manually labelling hundreds of thousands of aspect mentions. At the same time, it introduces a methodological limitation: ratings are useful proxies, but they do not always perfectly match the sentiment expressed in a local sentence.

This limitation is important for interpreting the later model comparison.

A passenger may give a low overall rating because of a delay while still writing positively about the crew. Another passenger may rate Value For Money poorly even if dissatisfaction is expressed indirectly rather than through explicit price-related language. Silver labels therefore allow the pipeline to operate at scale, but they also introduce noise into the supervision. The methodology treats this not as a minor technical detail, but as a central feature of the research design. The quality of the labels becomes part of the question: the comparison between classical and transformer-based models is also a test of how different models behave under imperfect supervision.

The lexicon baseline is included at this stage as a transparent comparison. It classifies sentiment using positive and negative words found near the aspect mention, with basic negation handling. Its strength is interpretability: the reason for the classification can be traced to visible words. Its weakness is that airline dissatisfaction is often expressed through events rather than adjectives. A passenger who writes “we waited on the tarmac for three hours” may not use an explicitly negative word, but the event clearly signals dissatisfaction. Comparing the lexicon baseline with TF-IDF therefore shows whether learning domain-specific textual patterns improves on a purely dictionary-based reading of passenger feedback.

The classification stage uses a time-based train/test split. Reviews before the selected cutoff date are used for training, while later reviews are reserved for testing. This is more realistic than a random split because airline reviews are temporal data. A random split could allow the model to learn from future patterns when predicting earlier or contemporary reviews. The time-based approach better reflects a real deployment setting, where a model is trained on past passenger feedback and then applied to future reviews. This design choice also supports a more credible evaluation of model generalisation.

The sixth module examines rating–aspect alignment. Once aspects have been extracted and sentiment has been classified, the analysis returns to the numerical ratings to understand which service dimensions are most closely associated with the overall passenger rating. Pearson correlations are computed between the overall rating and each service sub-rating. This step connects the ABSA pipeline to a managerial question: which dimensions matter most for the passenger’s final judgement? A service aspect may be frequently mentioned, but frequency alone does not mean that it drives satisfaction. The alignment analysis helps distinguish visibility from importance.

The seventh module introduces time into the analysis. Monthly average ratings are computed from the date-flown variable, and rolling averages are used to smooth short-term fluctuations. Major aviation events are then annotated on the timeline. This technique allows the analysis to move beyond static averages and observe how passenger satisfaction changes across industry conditions. Airline service does not operate in a stable environment. Pandemics, strikes, safety events, airspace closures, recovery periods, and technology failures can all influence passenger expectations, airline operations, and the emotional tone of reviews. The time-series module therefore situates individual feedback within a broader industry context.

The eighth module builds a predictive model of overall rating and studies delay thresholds. Ridge regression is used because the purpose is both predictive and interpretive. The model combines structured sub-ratings with contextual and textual variables such as traveller type, cabin class, seasonal indicators, COVID-period indicators, delay mentions, lexicon score, and review length. Ridge regression is appropriate because many of these predictors are correlated with each other. The regularisation term stabilises the estimates and makes the contribution of each variable easier to interpret. In this

way, the model is not only a prediction tool, but also a structured way to understand which information in the review record is most useful for explaining overall satisfaction.

The same module examines delay duration as a possible frustration threshold. Delay mentions are detected in the text, and when passengers provide explicit time information, approximate duration is extracted and grouped into delay bins. The purpose is to understand whether satisfaction decreases gradually as delays become longer or whether there is a point at which the passenger’s evaluation changes more sharply. This is managerially relevant because service recovery decisions often depend on timing. A short delay, a moderate delay, and a severe delay may require different communication and compensation strategies.

The ninth module performs anomalous review detection. The goal is not to declare reviews fake, but to identify observations that are unusual based on textual characteristics, while acknowledging that online-review authenticity is a broader platform-level issue (Luca and Zervas, 2016). The module uses content-based signals such as extreme brevity, low lexical diversity, title–body overlap, vague language, short unverified text, and outlier behaviour detected through TF-IDF features and Isolation Forest. Rating extremity is deliberately excluded as a suspicion signal, because very negative ratings are common in voluntary review platforms and may reflect legitimate severe experiences. The anomaly score therefore measures textual unusualness, not proven inauthenticity.

This distinction is important for both methodological and managerial reasons. Strong dissatisfaction should not automatically be treated as suspicious. A very negative review may be the most valuable kind of feedback if it describes a genuine service failure. The anomaly module therefore provides

a cautious screening tool rather than a judgement of authenticity. It helps identify records that may require additional inspection while preserving the integrity of legitimate extreme feedback.

The tenth module analyses routes and haul type. Passenger expectations change depending on the journey. A short domestic flight and an ultra-long-haul international flight are not evaluated according to the same standards. The module parses route information and maps journeys into haul categories such as short-haul, medium-haul, long-haul, and ultra-long-haul. This creates a route-level segmentation that allows satisfaction and sentiment patterns to be compared across travel distance. The managerial relevance is clear: a service feature that is secondary on a short flight may become essential on a long-haul journey.

The eleventh module performs Aspect–Opinion Pair Extraction. While the classifier predicts sentiment labels, this module tries to recover the local words that explain those labels. For each aspect mention, the surrounding text is searched for opinion words, and polarity is adjusted when negation is present. This creates pairs between service aspects and evaluative language. The value of this step is interpretability. It does not only say that passengers are negative about Cabin Comfort or positive about Cabin Crew Service; it shows the words through which those judgements are expressed.

This module is especially useful because it brings the analysis closer to the voice of the passenger. For management, a polarity score is informative, but the words behind the score are often more actionable. Knowing that passengers associate an aspect with words such as “rude”, “helpful”, “cramped”, “delayed”, or “expensive” helps translate quantitative results into service priorities. Aspect–opinion pairs therefore act as a bridge between numerical sentiment analysis and the concrete language of customer experience.

The twelfth module examines traveller type and cabin class interactions. Passenger expectations are not homogeneous. Business travellers may prioritise punctuality, reliability, and efficiency, while leisure travellers may care more about price, comfort, or the emotional quality of the trip. Families may experience boarding, seating, delays, and staff support differently from solo travellers. Cabin class also changes expectations. Economy passengers may evaluate the experience through affordability, while premium passengers judge the airline against a higher service promise. This module therefore examines whether satisfaction patterns differ across traveller groups and cabin classes.

The thirteenth module analyses aircraft type. Aircraft names in review data are often written inconsistently, so the module normalises them into broader aircraft families and categories such as narrowbody, widebody, regional jet, and turboprop. This makes it possible to compare satisfaction across aircraft types. The motivation is that aircraft type can influence comfort, entertainment, cabin environment, and perceived service quality. A widebody aircraft used on a long-haul route creates a different passenger experience from a narrowbody aircraft used on a short route, and the methodology preserves this operational context.

The fourteenth module studies weather mentions and attribution. The purpose is not to reconstruct actual meteorological conditions from external weather data, but to analyse whether passengers themselves mention weather when explaining their experience. This distinction is important because satisfaction depends not only on what happened, but also on how passengers interpret why it happened. Weather categories such as turbulence, precipitation, ice and de-icing, fog and visibility, wind and storm, and generic weather

language are identified in the text. The module then compares ratings between reviews with and without weather mentions, and between weather reviews with and without delay.

This stage is grounded in attribution theory. Passengers may respond differently to the same negative outcome depending on whether they see the cause as controllable or uncontrollable. A delay attributed to severe weather may be judged differently from a delay attributed to poor airline organisation (Weiner, 1985; Folkes, 1984). However, airlines still control communication, care, and recovery. This means that weather attribution is not only an operational issue, but also a communication and service-management issue.

The fifteenth module analyses peak season in a hemisphere-aware way. Seasonality is important in aviation, but it cannot be defined globally with a single calendar rule. June to August represents the summer peak in the Northern Hemisphere, while December to February represents the summer peak in much of the Southern Hemisphere. Applying one universal definition of summer would misclassify part of the data. The module therefore identifies hemisphere through route-related information and defines peak season separately for Northern and Southern Hemisphere travel. This makes the seasonal analysis more appropriate for a global airline corpus.

The sixteenth module applies a Difference-in-Differences design to major aviation events. A simple before-and-after comparison can be misleading because satisfaction is seasonal and because review behaviour changes over time. Difference-in-Differences compares the change around an event with the change observed in a comparable control period. The estimator is defined as:

$$\text{DiD} = (\bar{Y}_{\text{post,treatment}} - \bar{Y}_{\text{pre,treatment}}) - (\bar{Y}_{\text{post,control}} - \bar{Y}_{\text{pre,control}})$$

This design is used to estimate the impact of major aviation events while reducing the risk of confusing event effects with normal temporal variation. The interpretation remains cautious because the data are observational, not experimental. Nevertheless, the method provides stronger evidence than a simple comparison of ratings before and after an event, and it is particularly appropriate in a domain where seasonality and disruption are constant background forces (Card and Krueger, 1994).

The seventeenth module studies seasonal sensitivity by segment. The question is not only whether peak season changes satisfaction, but which passengers and journeys are most exposed to that change. The module compares peak-season effects across carrier type, cabin class, and haul type. This segmentation matters because seasonal pressure may not affect all business models or customer groups in the same way. Low-cost and legacy carriers may absorb demand pressure differently; premium passengers may react more strongly when service falls below expectations; long-haul routes may be more vulnerable to accumulated operational stress. This stage therefore turns seasonality from a general time pattern into a more actionable management question.

The eighteenth and final module fine-tunes DistilBERT for aspect sentiment classification. This module acts as the neural benchmark against the classical TF-IDF model. Following the aspect-aware logic of Sun et al. (2019), the model receives both the local review context and the target aspect name. The input format is:

[CLS] context [SEP] aspect name [SEP]

This structure tells the model which service dimension it should evaluate. The same review context can therefore be interpreted differently depending

on whether the target aspect is Punctuality, Cabin Comfort, Cabin Crew Service, Food & Beverages, or Value For Money. One `distilbert-base-uncased` classifier is fine-tuned per aspect using the same silver-label logic and time-based split as the TF-IDF model.

DistilBERT is included because it provides a practical test of modern contextual modelling. Transformer models are often expected to improve performance in language tasks where context matters. However, the methodological question here is not simply whether a larger model is more powerful, but whether additional model capacity is useful when the labels are derived from structured ratings rather than manual annotation. DistilBERT therefore functions as a benchmark for testing whether the limiting factor is representation power or supervision quality. This is an important distinction for applied analytics, because investing in a more complex model is not always the best solution if the main constraint is label noise.

The pipeline concludes by saving figures, tables, model metrics, intermediate datasets, and final analytical outputs. This output structure makes the workflow auditable. Each module produces evidence that can be inspected independently, while the full pipeline preserves the connection between raw passenger narratives and final interpretation. The methodology therefore follows a cumulative logic: reviews are cleaned, explored, structured into aspects, classified by sentiment, connected to ratings, analysed across operational and contextual dimensions, tested through causal and seasonal designs, and finally benchmarked against a transformer model. The result is not only a technical NLP workflow, but a decision-oriented framework for understanding how passengers evaluate airline service.

Chapter 6

Results

6.1 Experimental Setting and Implementation

The results presented in this chapter were produced through the full execution of the modular ABSA pipeline described in Chapter 5. The implementation was developed in Python and organised as a reproducible workflow in which each module produces intermediate outputs, figures, tables, and saved metrics. This structure is important because the analysis does not rely on a single model or a single result. Instead, the evidence is built progressively: the review corpus is first transformed into aspect spans, those spans are classified by sentiment, the predicted aspect-level evidence is compared with structured ratings, and the resulting outputs are then examined across passenger segments, operational contexts, temporal patterns, weather mentions, and major aviation events.

The experiments were run on the cleaned corpus of 144,934 airline passenger reviews, covering 578 airlines and flights from April 2012 to March 2026. The input data combined free-text review narratives, an overall 1–10 rating, seven 1–5 service sub-ratings, airline information, date flown, verification status, cabin class, traveller type, route, and aircraft information when available. This combination of textual, numerical, and contextual data

allowed the pipeline to treat passenger satisfaction not as a single outcome, but as a layered service judgement.

Table 3: Implementation and experimental configuration.

Component	Configuration
Programming language	Python 3.10 or later.
Core libraries	pandas, NumPy, Matplotlib, Seaborn, scikit-learn, and SciPy.
Transformer libraries	PyTorch, Hugging Face Transformers, and Hugging Face Datasets.
Repository structure	Modular pipeline coordinated by <code>main.py</code> , with shared constants, paths, lexicons, and aspect definitions stored in <code>config.py</code> .
Data source	SkyTrax / airlinequality.com airline passenger reviews.
Corpus size	144,934 usable reviews after cleaning.
Aspect extraction	Rule-based BIO sequence tagging using domain-specific aspect triggers.
Classical ASC model	TF-IDF vectorisation with unigrams and bigrams, followed by Logistic Regression.
TF-IDF settings	Maximum 15,000 features, minimum document frequency of 2, and sublinear term-frequency scaling.
Classifier settings	Logistic Regression with L2 regularisation, $C = 1.0$, class balancing, and maximum 1,000 iterations.
Silver labels	Derived from structured sub-ratings where available; overall rating used as fallback where no direct sub-rating exists.
Predictive model	Ridge regression with standardised predictors and a time-based train/test split.
DistilBERT model	<code>distilbert-base-uncased</code> , fine-tuned separately for each aspect.
DistilBERT input format	Local review context combined with the target aspect name.
DistilBERT settings	Maximum sequence length of 128 tokens, batch size of 16, three training epochs, AdamW optimiser, learning rate 2×10^{-5} , weight decay of 0.01, and linear warmup over 10% of training steps.
Hardware execution	PyTorch used CUDA GPU acceleration when available, with CPU fallback otherwise. The exact GPU model was not stored in the repository outputs.
Output format	Figures, tables, CSV metrics, model checkpoints, and intermediate analytical datasets saved to the project output directory.

Table 3 summarises the experimental configuration behind the results. The table should be read as the technical context of the chapter rather than as an additional methodological contribution. The most important element is that the pipeline combines interpretable classical methods with a transformer

benchmark. The TF-IDF and Logistic Regression model provides a strong, transparent baseline for aspect sentiment classification, while DistilBERT tests whether a modern contextual model improves performance when the labels are derived automatically from structured ratings.

A second important feature of the experimental setting is the distinction between gold labels and silver labels. No manually annotated gold-standard aspect-sentiment dataset exists for this corpus. Instead, sentiment labels are derived automatically from the structured passenger ratings. This allows the analysis to operate at scale, but it also introduces noise. The results should therefore be interpreted with this in mind: the models are not only being evaluated on how well they understand text, but also on how well they can learn from imperfect rating-derived supervision.

The remainder of the chapter presents the results in the same order as the pipeline. The first part focuses on the core ABSA task: aspect extraction, aspect sentiment classification, model comparison, and rating–aspect alignment. The second part moves from the model outputs to the managerial interpretation of passenger satisfaction across carrier type, cabin class, time, route, aircraft, traveller segment, weather, seasonality, and aviation events.

6.2 Core ABSA Results and Model Performance

The first result concerns aspect extraction. The BIO tagger identified 717,904 aspect spans across 138,813 reviews, corresponding to 95.8% review coverage. This means that almost all reviews contained at least one recognised service aspect. The high coverage is important because it confirms that the aspect schema captures the main vocabulary passengers use when discussing airline experiences.

Table 4: Aspect span counts from BIO tagging.

Aspect	Spans	% of total
Cabin Crew Service	213,723	29.8%
Cabin Comfort	149,795	20.9%
Value For Money	125,667	17.5%
Punctuality	115,100	16.0%
Food & Beverages	113,619	15.8%
Total	717,904	100.0%

Table 4 shows how the detected aspect spans are distributed across the five service dimensions. Cabin Crew Service is the most frequent aspect, representing 29.8% of all spans. This means that passengers very often describe the airline through its human service interface: crew behaviour, helpfulness, friendliness, rudeness, or professionalism. Cabin Comfort is the second most frequent aspect, followed by Value For Money, Punctuality, and Food & Beverages. The distribution is useful because it shows what passengers choose to talk about when they write reviews. It does not yet tell us which aspects drive satisfaction most strongly, but it tells us which parts of the journey are most visible in the passenger narrative.

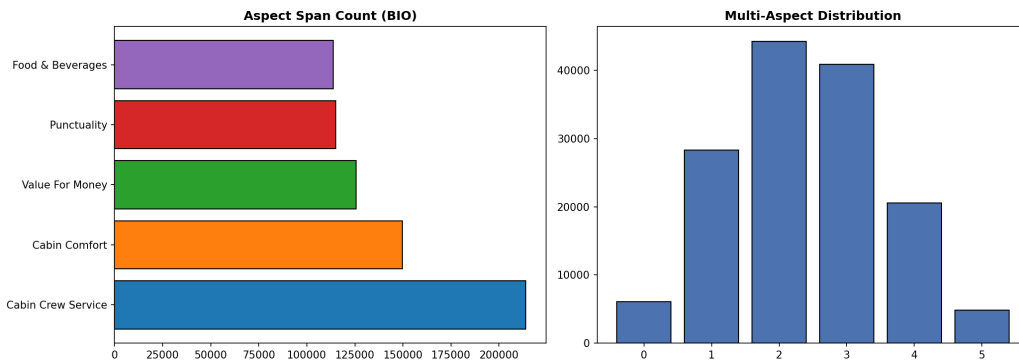


Figure 5: Aspect span counts and multi-aspect review distribution derived from BIO tagging.

Figure 5 should be read in two parts. The first part shows the number of spans detected for each aspect, making the dominance of Cabin Crew Service immediately visible. The second part shows how many distinct aspects appear in each review. This is especially important because 76.2% of reviews contain two or more distinct aspects. The figure therefore provides visual evidence for the central premise of the study: airline reviews are usually multi-aspect texts. A passenger rarely evaluates only one dimension of the journey. The same review may discuss delay, staff, seat comfort, food, and price. This validates the use of ABSA rather than document-level sentiment analysis.

After the aspect spans are extracted, the next step is aspect sentiment classification. The classical model uses TF-IDF and Logistic Regression, while the simplest comparison point is a lexicon baseline. The purpose of this comparison is to understand whether a model trained on airline-specific text improves over a transparent dictionary-based approach.

Table 5: TF-IDF + Logistic Regression ASC performance.

Model / Aspect	Precision	Recall	F1	Accuracy
Lexicon Baseline	0.557	0.494	0.462	—
TF-IDF + LogReg overall	0.607	0.611	0.608	0.611
Punctuality	0.621	0.625	0.622	0.625
Cabin Comfort	0.608	0.610	0.608	0.610
Cabin Crew Service	0.603	0.607	0.603	0.607
Food & Beverages	0.617	0.621	0.618	0.621
Value For Money	0.583	0.588	0.584	0.588

Table 5 reports precision, recall, F1-score, and accuracy for the TF-IDF classifier. Precision measures how reliable the model predictions are; recall measures how much of each class the model is able to recover; F1-score balances precision and recall. Because the task has three sentiment classes,

macro-F1 is especially important because it treats negative, neutral, and positive sentiment equally rather than allowing the largest class to dominate the evaluation. The TF-IDF model achieves an overall macro-F1 of 0.608, compared with 0.462 for the lexicon baseline. This improvement shows that learning from the corpus captures patterns that a simple sentiment dictionary misses.

The aspect-level rows show that performance is relatively stable across service dimensions, but not identical. Punctuality and Food & Beverages reach the highest F1-scores, while Value For Money is the most difficult aspect. This pattern is meaningful. Punctuality often contains concrete event vocabulary such as delays, cancellations, missed connections, or late departures. Food & Beverages also tends to contain more direct evaluative language. Value For Money, by contrast, is more abstract and context-dependent. A word such as “cheap” can be positive when it means good value, but negative when it suggests poor quality. This makes the aspect harder to classify with surface text features.

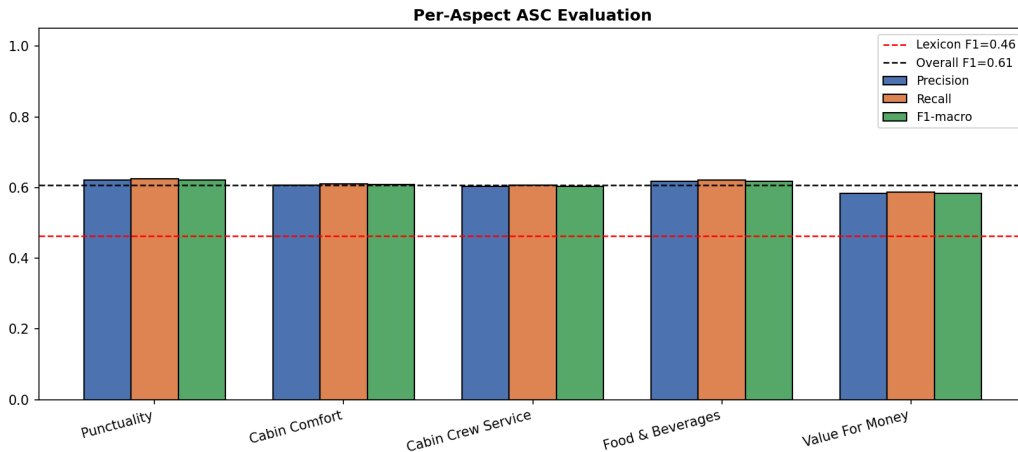


Figure 6: Per-aspect evaluation metrics for the TF-IDF Aspect Sentiment Classification model.

Figure 6 visualises the information in Table 5. Each aspect is represented by grouped bars for precision, recall, and F1-score, while the horizontal reference lines show the overall model level and the lexicon baseline. The figure is useful because it allows the reader to see immediately that the TF-IDF model is consistently above the lexicon baseline across all aspects. The chart also shows that no single aspect completely dominates the model performance. The classifier is not excellent on one category and weak on all others; rather, it provides a stable improvement across the full aspect schema.

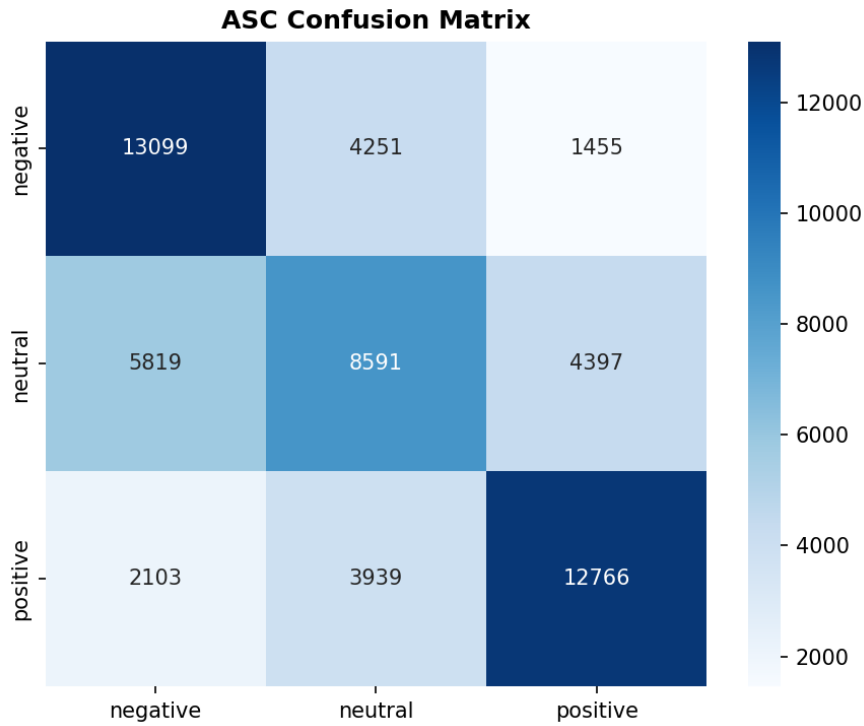


Figure 7: Overall confusion matrix for the TF-IDF Aspect Sentiment Classification model.

Figure 7 shows the confusion matrix for the TF-IDF classifier. A confusion matrix should be read by comparing the true labels on one axis with

the predicted labels on the other. The diagonal cells represent correct predictions, while the off-diagonal cells show where the model confuses one sentiment class with another. The figure is useful because it reveals not only how often the model is correct, but also what type of mistakes it makes. In this task, the neutral class is particularly important because neutral or mixed sentiment is harder to identify than clearly positive or clearly negative language. The confusion matrix therefore helps explain why macro-F1 is a better summary metric than accuracy alone.

The next model is fine-tuned DistilBERT. This experiment tests whether a contextual transformer improves performance over the classical TF-IDF representation. The transformer receives the local review context together with the aspect name, so the model is not only reading the text, but also being told which service dimension it should evaluate.

Table 6: Fine-tuned DistilBERT ASC performance with per-class F1.

Aspect	F1	Accuracy	F1 neg.	F1 neu.	F1 pos.	Test n
Punctuality	0.551	0.655	0.751	0.145	0.757	11,887
Cabin Comfort	0.593	0.648	0.749	0.308	0.722	9,629
Cabin Crew Service	0.590	0.660	0.729	0.263	0.777	15,725
Food & Beverages	0.619	0.694	0.808	0.317	0.732	7,723
Value For Money	0.520	0.662	0.790	0.165	0.604	10,916
Overall	0.574	0.664				55,880

Table 6 reports the DistilBERT results. The table should be read carefully because it contains both overall aspect performance and per-class F1-scores. The model performs well on negative and positive classes, but struggles with the neutral class. This is visible in the low neutral F1 values, especially for Punctuality and Value For Money. The result suggests that DistilBERT can recognise strong sentiment signals, but has difficulty separating genuine neutrality from weakly positive, weakly negative, or mixed cases. This is

important because neutral sentiment in airline reviews is rarely simple. It often reflects a compromise between acceptable and disappointing parts of the journey.

The overall DistilBERT macro-F1 is 0.574. Food & Beverages is the best aspect for DistilBERT, while Value For Money is the weakest. Again, this pattern reflects the difference between concrete and abstract service dimensions. Food-related sentiment tends to be expressed through more direct evaluative language, while Value For Money depends on expectations, price fairness, and the perceived relationship between cost and service quality.

Table 7: Three-way ASC model comparison.

Aspect	Lexicon	TF-IDF	DistilBERT	TF-IDF gain	BERT gap
Punctuality	0.462	0.622	0.551	+0.160	-0.071
Cabin Comfort	0.462	0.608	0.593	+0.146	-0.015
Cabin Crew Service	0.462	0.603	0.590	+0.141	-0.013
Food & Beverages	0.462	0.618	0.619	+0.156	+0.001
Value For Money	0.462	0.584	0.520	+0.122	-0.064
Overall	0.462	0.608	0.574	+0.146	-0.034

Table 7 brings the three models together. The first comparison is between the lexicon baseline and TF-IDF. TF-IDF improves over the lexicon baseline on every aspect, confirming that domain-specific learning is useful. The second comparison is between TF-IDF and DistilBERT. DistilBERT underperforms TF-IDF on four of five aspects, with approximate parity only on Food & Beverages. The overall gap is not enormous, but it is directionally important because it contradicts the common expectation that a transformer model will automatically outperform a classical feature-based classifier.

This result should not be read as evidence that DistilBERT is a weak model. Instead, it points to the quality of supervision. The labels are silver

labels derived from ratings, not manually annotated aspect-level labels. A high-capacity model may therefore learn inconsistencies in the labels rather than extracting more meaningful linguistic patterns. The TF-IDF model, being simpler and more regularised, appears more robust under this kind of noisy supervision.

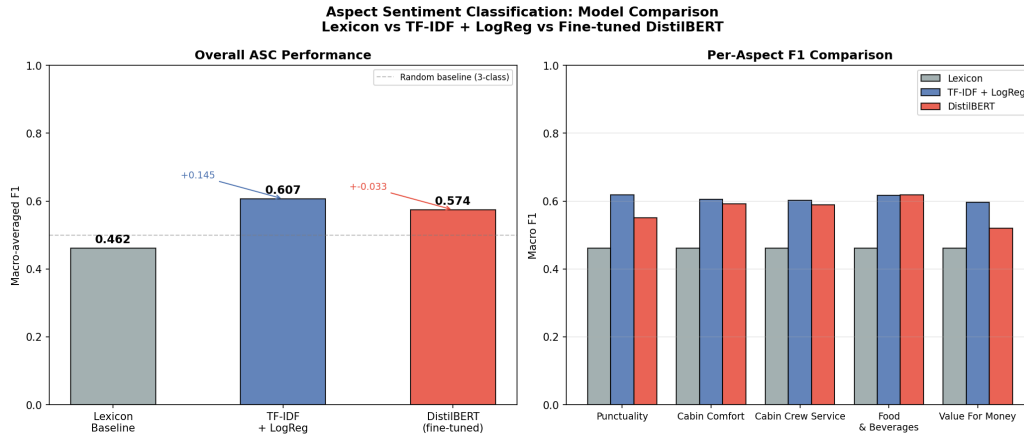


Figure 8: Three-way ASC model comparison: Lexicon vs. TF-IDF vs. DistilBERT.

Figure 8 makes the model comparison easier to interpret visually. The left side summarises overall macro-F1 across the three approaches, while the right side compares the models aspect by aspect. The useful information in this figure is the shape of the comparison: the lexicon baseline is consistently lowest, TF-IDF is consistently strong, and DistilBERT does not create a clear improvement despite being more sophisticated. For management, this has a practical implication. More complex AI is not always better if the input labels are noisy. In this case, improving label quality may be more valuable than moving immediately to larger models.

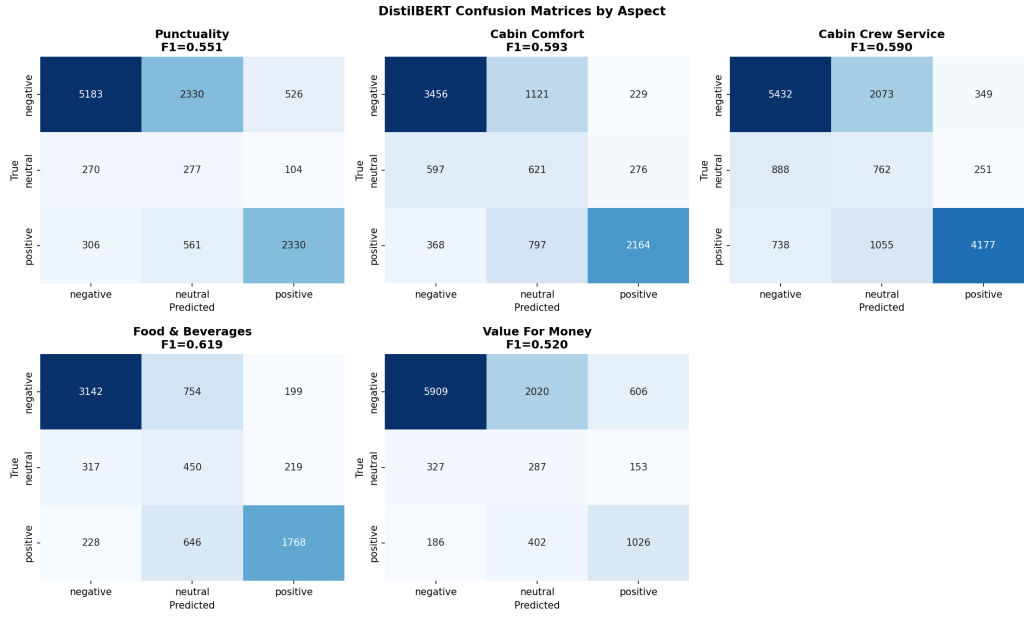


Figure 9: DistilBERT confusion matrices by aspect, illustrating the neutral class bottleneck.

Figure 9 shows one confusion matrix for each aspect in the DistilBERT experiment. These matrices should be read in the same way as Figure 7: correct classifications appear on the diagonal, while off-diagonal cells represent errors. The figure is especially useful because it shows that the problem is not distributed evenly across all classes. The neutral class is the bottleneck. DistilBERT often recognises clearly negative or clearly positive cases, but neutral cases are frequently absorbed into neighbouring classes. This is consistent with the nature of airline reviews, where neutral ratings may reflect mixed experiences rather than truly neutral language.

Table 8: Lexicon–TF-IDF prediction disagreement rate by aspect.

Aspect	Disagreement rate
Value For Money	59.8%
Food & Beverages	50.1%
Punctuality	45.8%
Cabin Comfort	44.4%
Cabin Crew Service	38.1%

Table 8 compares the predictions of the lexicon baseline with those of the TF-IDF model. The disagreement rate is highest for Value For Money, at 59.8%, and lowest for Cabin Crew Service, at 38.1%. This table is useful because it identifies which aspects are most difficult for simple sentiment dictionaries. Cabin Crew Service often contains direct sentiment vocabulary, such as “friendly”, “rude”, “helpful”, or “attentive”. Value For Money is more context-dependent. Price-related words are not automatically positive or negative; their meaning depends on the perceived fairness of the exchange. The disagreement table therefore shows where domain-aware modelling adds the most value.

The next step connects aspect-level evidence with overall passenger satisfaction. The pipeline therefore returns to the structured sub-ratings and asks which service dimensions align most strongly with the overall score.

Table 9: Pearson correlation between sub-ratings and overall rating.

Dimension	Pearson r
Value For Money	0.899
Ground Service	0.853
Cabin Staff Service	0.794
Seat Comfort	0.788
Food & Beverages	0.788
Wifi & Connectivity	0.711
Inflight Entertainment	0.695

Table 9 shows the Pearson correlation between each sub-rating and the overall rating. Correlation values close to 1 indicate that the sub-rating moves very closely with the final passenger judgement. The strongest dimension is Value For Money, with $r = 0.899$. This is a major finding because it means that passengers' global evaluation is highly aligned with whether they feel the flight was worth the price paid. Ground Service is the second strongest dimension, followed by Cabin Staff Service, Seat Comfort, and Food & Beverages. The lowest correlations belong to Wifi & Connectivity and Inflight Entertainment.

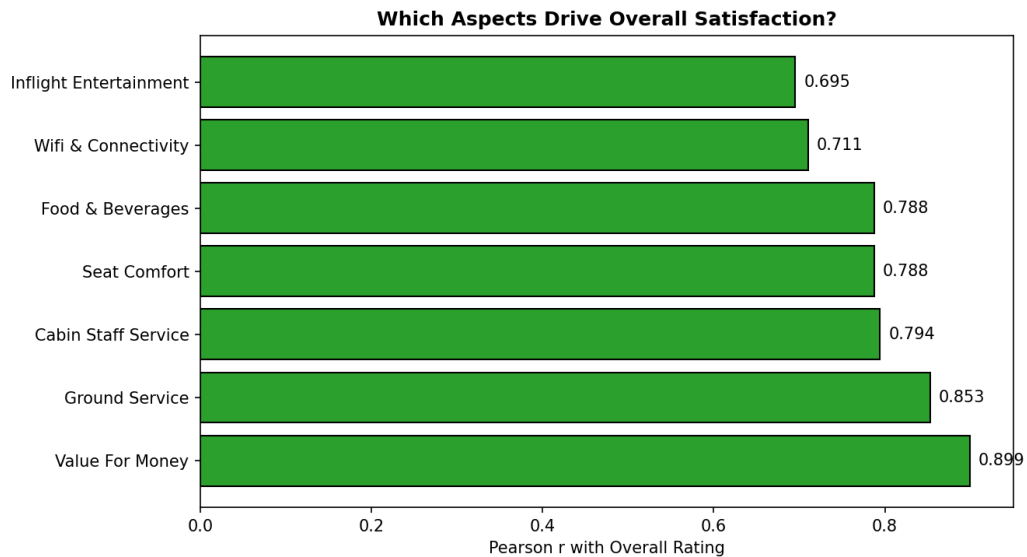


Figure 10: Pearson correlation between individual aspect sub-ratings and overall passenger satisfaction.

Figure 10 visualises the same ranking as a horizontal bar chart. The figure should be read from the longest bar to the shortest. The message is clear: Value For Money is not simply one aspect among others; it is the strongest structured predictor of overall satisfaction. This does not mean that comfort, staff, or food are unimportant. Rather, it means that passengers interpret these elements through the wider lens of fairness and exchange. If the service

feels poor relative to the fare paid, the overall score suffers strongly. From a managerial point of view, this figure suggests that pricing transparency, expectation management, and perceived fairness are central to satisfaction.

6.3 Passenger Satisfaction, Contextual Effects, and Managerial Evidence

The next set of results moves beyond the core ABSA task and uses the model outputs to interpret passenger satisfaction from a managerial perspective. The section follows a progressive logic. It first identifies the strongest satisfaction drivers, especially Value For Money, before examining how satisfaction changes across airline business models and cabin classes. It then moves from product context to journey context by considering delay, route length, traveller type, and aircraft category. Finally, it examines external and temporal conditions that shape passenger evaluations, including weather attribution, peak-season pressure, and major aviation disruption events.

This order is important because passenger satisfaction is not produced by one isolated factor. It begins with the perceived fairness of the exchange, is shaped by the service promise of the airline and cabin class, is affected by operational reliability during the journey, and is then reinterpreted through external causes such as weather, seasonality, and industry shocks. The aim of this section is therefore not only to report separate figures, but to show how the ABSA pipeline turns individual review signals into a layered explanation of airline passenger satisfaction.

Because Section 6 has already shown that Value For Money is the strongest structured correlate of the overall rating, the first result in this section extends that finding through predictive modelling. This creates a bridge between the model evaluation results and the managerial interpretation that

follows. The predictive model then examines which variables are most useful for explaining overall ratings. Ridge regression is used because it can handle correlated predictors while still providing interpretable coefficients. The model achieves $R^2 = 0.874$ and $\text{RMSE} = 1.162$ on the held-out post-2023 test set, indicating strong predictive performance.

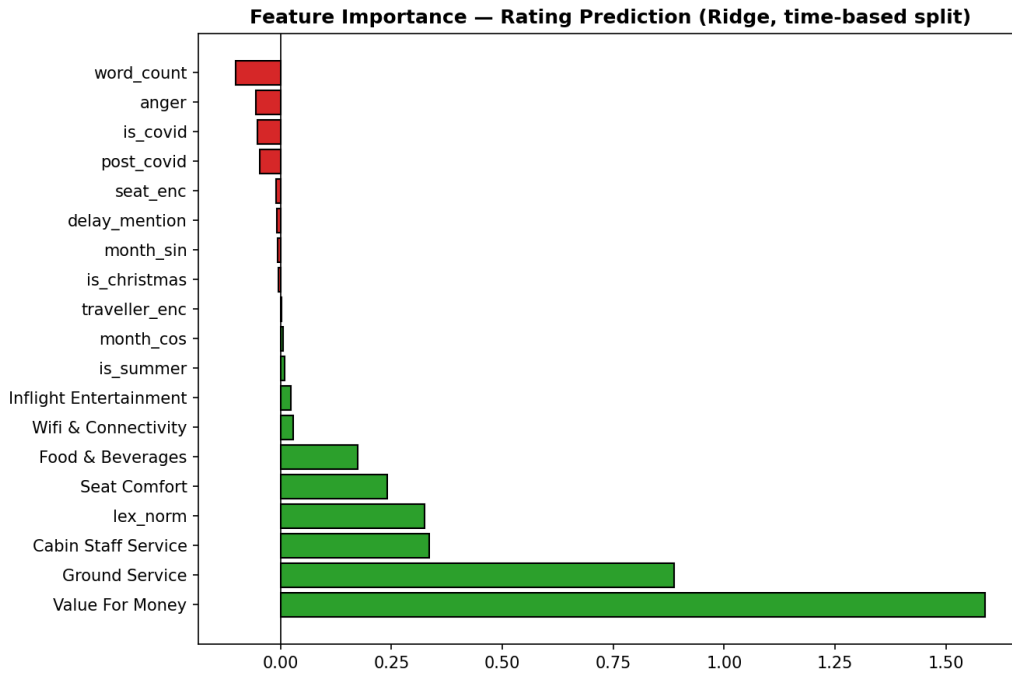


Figure 11: Ridge regression feature importance for rating prediction, highlighting Value For Money as the dominant predictor.

Figure 11 shows the Ridge regression coefficients. The bars indicate the direction and relative strength of each predictor. Positive bars are associated with higher overall ratings, while negative bars are associated with lower ratings. The most important message is the dominance of Value For Money. This confirms the earlier correlation result using a different analytical framework. The useful managerial information is that perceived value is not only correlated with satisfaction; it remains highly important even when other

ratings and contextual variables are considered together.

After identifying the main satisfaction driver, the analysis turns to airline business model. The first comparison examines European low-cost carriers and legacy airlines. The European subset contains 32,994 reviews, including 11,093 low-cost carrier reviews and 21,901 legacy carrier reviews. Legacy carriers record a slightly higher mean rating than low-cost carriers, with a statistically significant difference. At the same time, Value For Money is almost identical between the two groups, which suggests that passengers calibrate their expectations according to the business model. Low-cost passengers may tolerate fewer services if the fare is perceived as fair, while legacy passengers expect more because the service promise is higher.

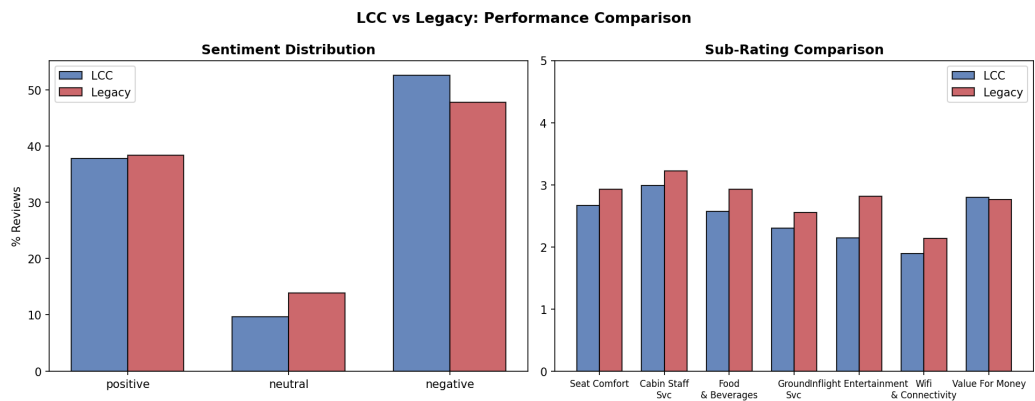


Figure 12: Performance comparison between Low-Cost Carriers (LCC) and Legacy airlines across sentiment distributions and sub-ratings.

Figure 12 contains two complementary views. The first compares sentiment distributions between LCC and legacy carriers; the second compares average sub-ratings. The useful information is not only whether one group performs better overall, but where the differences appear. Legacy carriers tend to perform better on service-related dimensions, while Value For

Money remains close between groups. This figure therefore tells a more nuanced story than a simple LCC-versus-legacy comparison. Passengers do not judge all airlines against the same standard; they evaluate the experience relative to the promise and price of the carrier.

The next product-level comparison is cabin class. Cabin class provides another important lens. The airline product is not uniform across Economy, Premium Economy, Business, and First Class. Each cabin class carries a different price, service promise, expectation level, and likelihood of public reviewing.

Table 10: Mean ratings by cabin class. Sub-ratings use a 1–5 scale; overall rating uses a 1–10 scale.

Dimension	Economy	Prem. Eco.	Business	First
Overall	4.06	—	6.16	—
Seat Comfort	2.59	2.91	3.51	3.51
Cabin Staff Service	2.86	3.09	3.77	3.60
Food & Beverages	2.58	2.76	3.42	3.17
Ground Service	2.25	2.68	3.20	3.11
Inflight Entertainment	2.48	2.86	3.25	3.07
Wifi & Connectivity	1.83	2.24	2.76	2.69
Value For Money	2.51	2.69	3.34	3.08

Table 10 shows that Business Class performs better than Economy across all service dimensions. This is expected because Business Class offers more space, better service, and a stronger product promise. However, the table also shows that premium service does not eliminate dissatisfaction. Even in higher cabins, some dimensions remain far from the maximum score. Wifi & Connectivity is the weakest dimension in every cabin class, and Economy Wifi & Connectivity, at 1.83, is the lowest sub-rating in the table. This indicates that connectivity is not merely an Economy problem, but a structural weakness across the airline product.

The table should also be interpreted with attention to review volume. Premium cabins contain fewer seats, and premium passengers may only write when the experience violates high expectations. As a result, Business and First Class reviews may disproportionately represent disappointed premium passengers. Their scores are higher than Economy, but they may still underestimate the satisfaction of the average premium traveller.

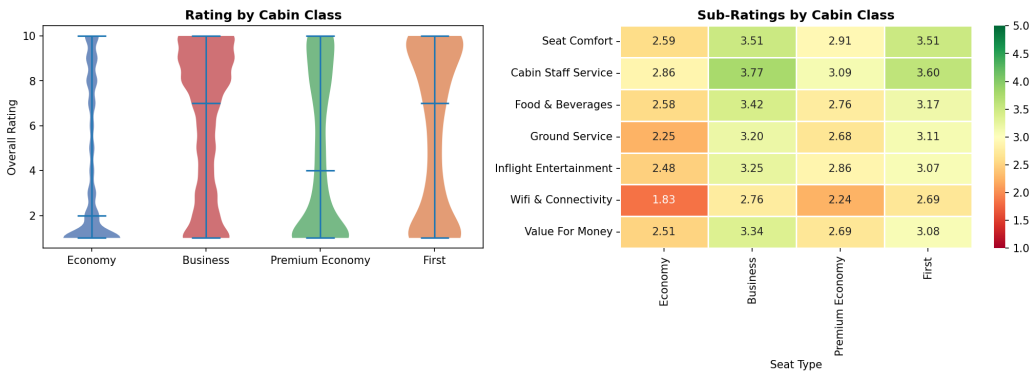


Figure 13: Overall rating distributions and sub-rating heatmaps by cabin class.

Figure 13 combines rating distributions and sub-rating heatmaps. The distribution plots show how ratings are spread within each cabin class, while the heatmaps show the relative strength and weakness of each service dimension. The figure is useful because it reveals both level and structure. Business Class generally has higher ratings, but the heatmap also shows that not every service dimension improves equally. For management, the figure suggests that cabin class upgrades do not simply raise satisfaction uniformly; they change the expectation structure of the passenger.

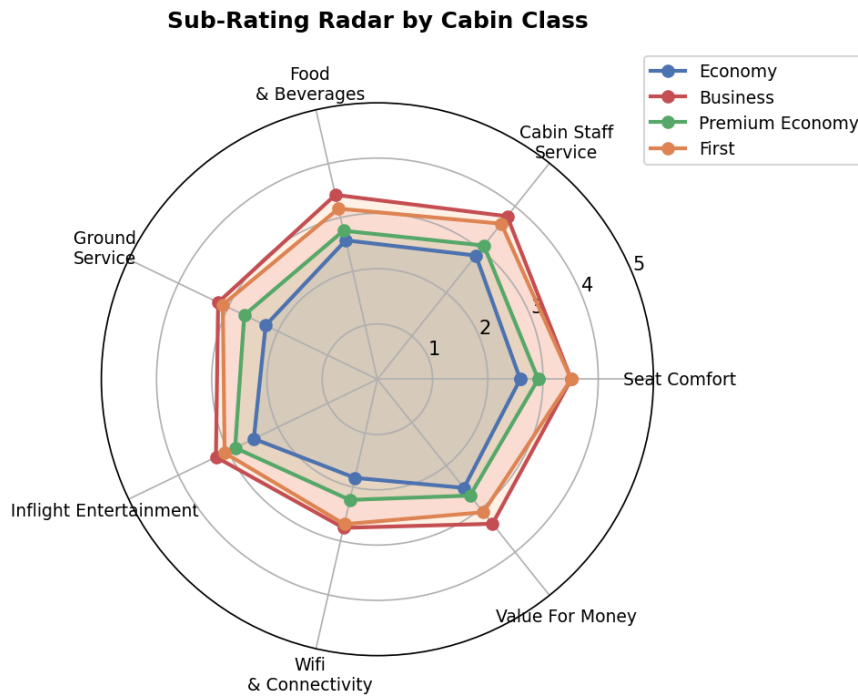


Figure 14: Radar chart comparing service dimension sub-ratings across the four cabin classes.

Figure 14 presents cabin-class performance as a radar profile. Each axis represents a service dimension, and each line represents a cabin class. The further the line extends outward, the higher the average sub-rating. The radar chart is useful because it shows the shape of the service experience rather than only the average level. Business and First Class generally extend further outward, while Economy remains closer to the centre. However, Wifi & Connectivity remains pulled inward across all classes. This visually reinforces the idea that some weaknesses persist even when passengers pay for a more premium product.

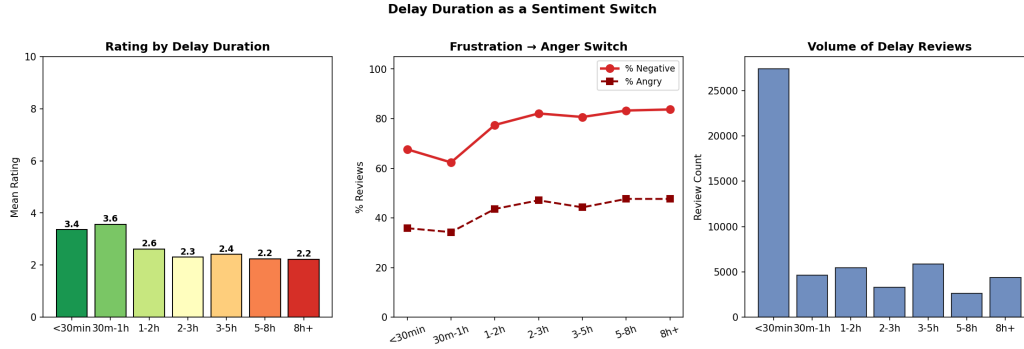


Figure 15: Delay duration as a sentiment switch, illustrating the frustration threshold and review volume by delay bin.

Operational reliability is then examined directly through delay duration. Figure 15 examines delay duration as a possible sentiment switch. The first panel shows mean rating by delay bin, the second shows the share of negative or angry language, and the third shows review volume. The important pattern is that ratings drop sharply between the one-hour and two-hour bins, then plateau at a low level. This suggests a satisfaction floor: once delay becomes severe enough, additional waiting time does not change the evaluation as much because the passenger has already interpreted the experience as a serious failure. For management, the implication is that intervention must happen before the experience reaches this collapse point. Communication, rebooking, compensation, and care become especially important before frustration becomes irreversible.

Route and haul type introduce another operational layer. The same service feature can have different importance depending on whether the passenger is taking a short domestic flight or an ultra-long-haul international journey.

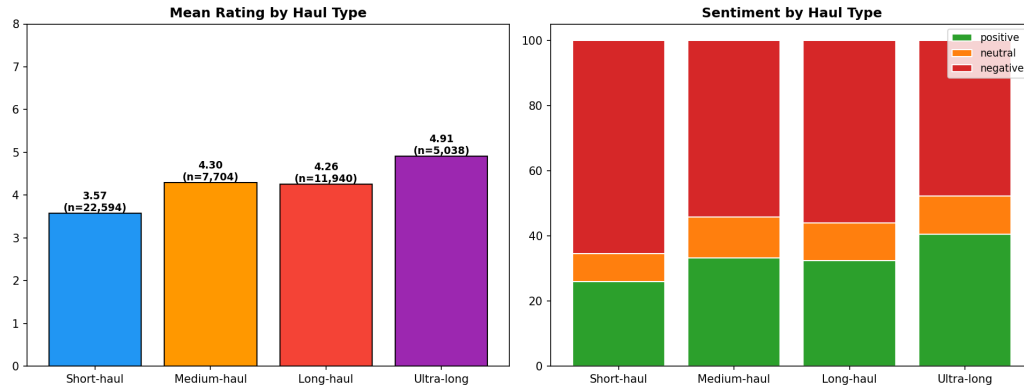


Figure 16: Mean overall rating and sentiment distribution across short, medium, long, and ultra-long-haul routes.

Figure 16 compares mean ratings and sentiment distributions across haul categories. The figure should be read by comparing both the height of the rating bars and the composition of positive, neutral, and negative sentiment. Short-haul routes record lower satisfaction than medium-, long-, and ultra-long-haul routes, while the share of positive sentiment increases with distance. This may reflect differences in product quality, passenger expectations, and review motivation. Long-haul flights often include a more complete service product, while short-haul flights may be more exposed to congestion, low-cost operating models, and limited onboard service. The figure shows that route context matters: satisfaction is not independent of journey type.

The traveller-type analysis then examines whether satisfaction differs across passenger groups and cabin classes.

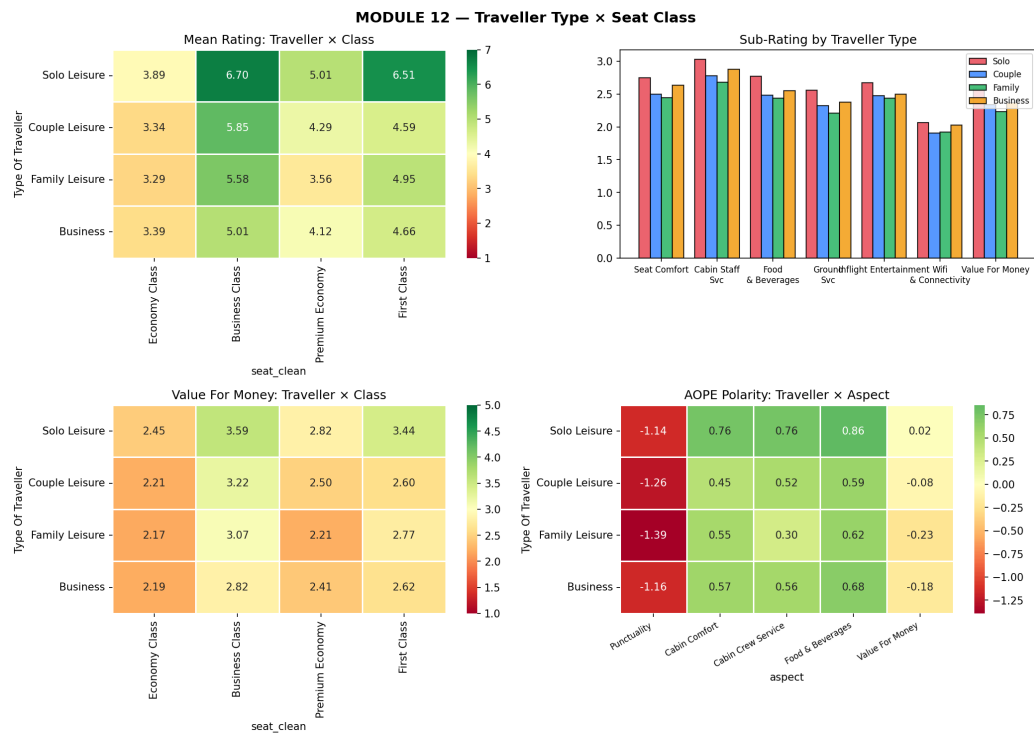


Figure 17: Interaction effects between traveller type, cabin class, and service aspect dimensions.

Figure 17 combines traveller type, cabin class, and service ratings. The figure should be read as an interaction plot: it does not only ask which group is more satisfied, but how satisfaction changes when traveller type and cabin class are considered together. Solo Business travellers in Business Class record the highest satisfaction, while Family Leisure passengers in Economy record the lowest. This pattern is understandable from a service-management perspective. Business Class provides a stronger service product, while families travelling in Economy may face more friction around seating, boarding, delays, and support needs.

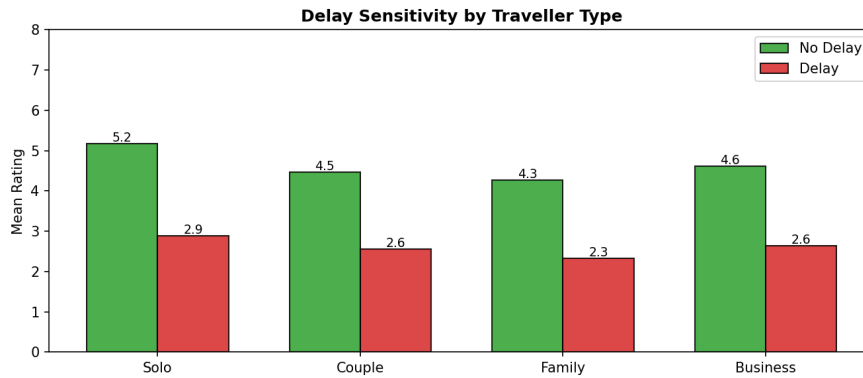


Figure 18: Delay convergence effect: all traveller groups converge to lower satisfaction when a delay is mentioned, regardless of baseline expectations.

Figure 18 adds delay to the traveller-type analysis. The figure shows that when delay is mentioned, satisfaction converges downward across all traveller groups. This is one of the clearest managerial results in the chapter. Different passengers begin with different expectations, but delay reduces those differences. Operational disruption creates a shared dissatisfaction floor. This means that delay management is not only important for one segment; it affects the entire customer base.

Aircraft type provides another product-level explanation of satisfaction.

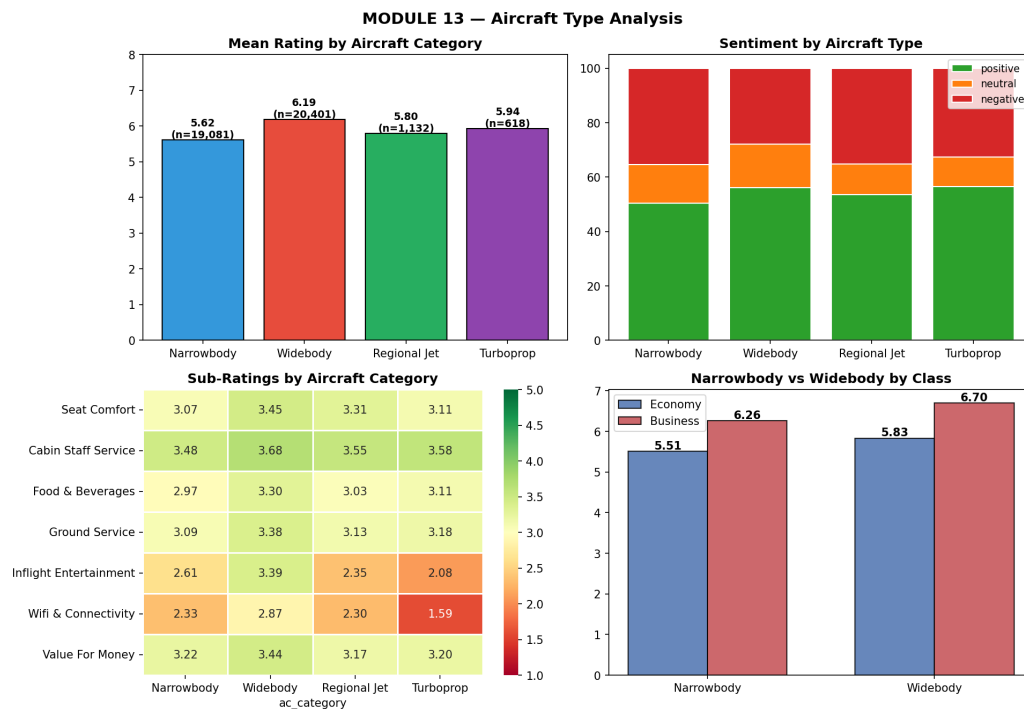


Figure 19: Satisfaction and sentiment breakdown comparing Widebody, Narrowbody, Regional Jet, and Turboprop aircraft.

Figure 19 compares satisfaction across aircraft categories. The figure combines mean ratings, sentiment distribution, and service sub-ratings. Widebody aircraft perform better than narrowbody aircraft, with the largest gap appearing in Inflight Entertainment. This is consistent with the product reality of airline operations: widebody aircraft are more common on long-haul routes and often provide a more complete onboard product. The figure shows that satisfaction is not shaped only by airline brand or staff behaviour; it is also influenced by the physical and technological environment in which the passenger travels.

Aspect–Opinion Pair Extraction then moves closer to the actual language used by passengers. Instead of only assigning a sentiment label, the module identifies the opinion words that appear near each aspect.

Table 11: Mean AOPE polarity by aspect.

Aspect	Mean polarity	Interpretation
Punctuality	-1.03	Only negative aspect
Value For Money	+0.17	Near-neutral / ambivalent
Cabin Comfort	+0.65	Moderately positive
Cabin Crew Service	+0.79	Positive
Food & Beverages	+0.80	Positive

Table 11 reports mean polarity scores. Negative values indicate more negative local opinion language, while positive values indicate more positive local opinion language. The table is useful because it separates the emotional tone of the words associated with each aspect. Punctuality is the only aspect with negative mean polarity, at -1.03. This means that when punctuality appears in review language, it is usually connected to negative expressions. Value For Money is close to neutral, suggesting ambivalence. Passengers may see good value in some cases and poor value in others, and the polarity depends heavily on context. Cabin Comfort, Cabin Crew Service, and Food & Beverages are positive on average.

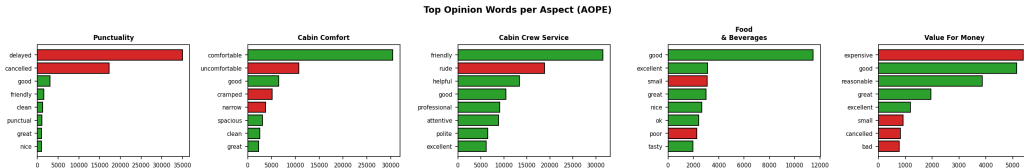


Figure 20: Top opinion words associated with each extracted service aspect.

Figure 20 shows the most common opinion words associated with each service aspect. This figure is important because it translates the numerical results back into passenger language. The most common opinion word is “delayed”, appearing 37,619 times, more often than “friendly” or “good”. This confirms that punctuality failures are not only frequent operational

events; they also dominate the emotional vocabulary of airline reviews. For management, this type of output is actionable because it shows the words passengers repeatedly use when describing each service dimension.

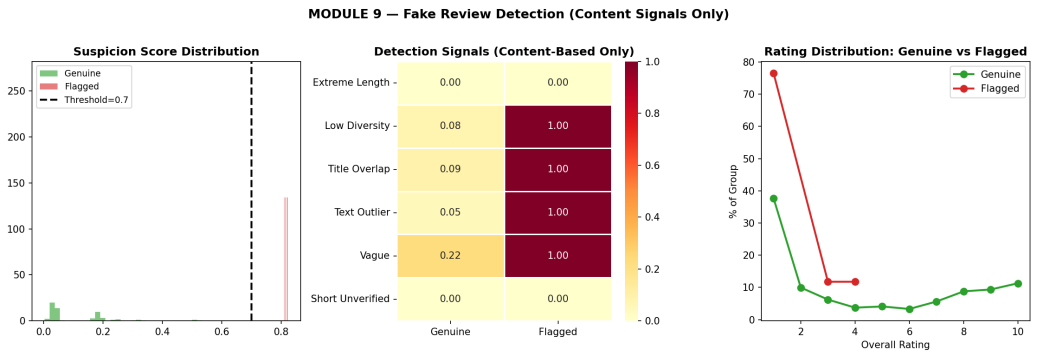


Figure 21: Anomalous review detection outputs based exclusively on content signals.

Before moving to external contextual effects, the analysis includes a content-based anomaly check. Figure 21 summarises the anomalous review detection module. The figure should not be read as a fake-review count. Instead, it shows which reviews are unusual according to content-based signals such as brevity, low lexical diversity, title–body overlap, vagueness, and outlier behaviour. At the selected threshold, 4,941 reviews are flagged, corresponding to 3.4% of the corpus. The useful information is cautionary: extreme dissatisfaction is not treated as suspicious by itself. This is important because in airline reviews, genuine severe failures often produce very negative ratings. The module therefore identifies textual unusualness, not confirmed manipulation.

After the product and journey-level analyses, the results move to the temporal structure of satisfaction. The time-series analysis places passenger satisfaction within the broader history of the airline industry. This is important because the period covered by the corpus includes major shocks: the

COVID-19 pandemic, the post-pandemic recovery, strikes, safety concerns, technology failures, and changes in passenger demand.

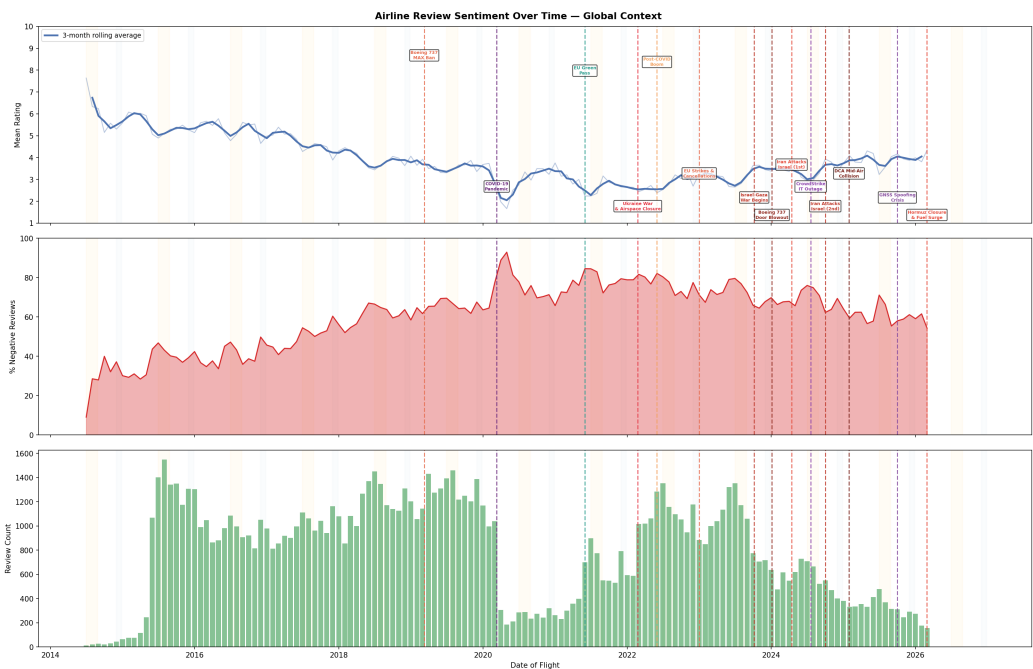


Figure 22: Longitudinal airline review sentiment over time, annotated with major global aviation disruption events.

Figure 22 should be read as a story over time. The main line shows how the mean overall rating changes month by month, while the annotations mark major aviation events. The lower panels show the volume of reviews and the sentiment composition over time. The useful information is the long-term pattern: satisfaction declines from the earlier years of the corpus, reaches very low levels around the pandemic and post-pandemic disruption period, and then partially recovers. This figure shows that passenger satisfaction is not only a matter of individual service encounters. It is also shaped by the operational and emotional climate of the airline industry.

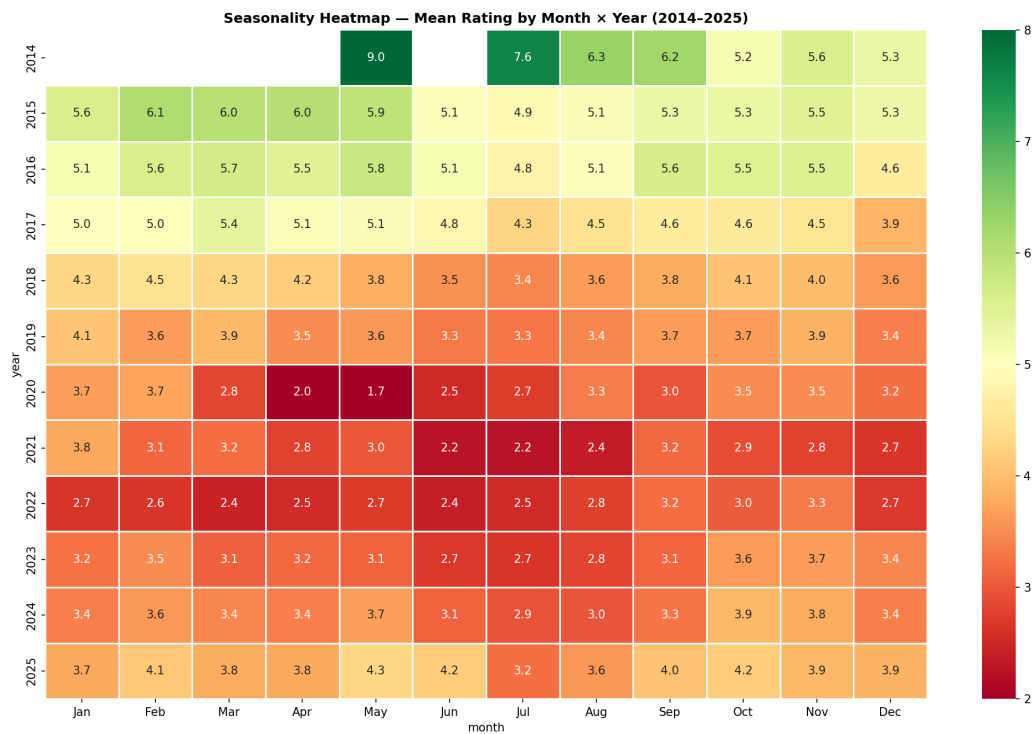


Figure 23: Seasonality heatmap displaying mean overall rating by month and year.

Figure 23 displays mean ratings by month and year. A heatmap should be read by comparing colour intensity across rows and columns. Each cell represents a month in a specific year, and the colour shows whether ratings were higher or lower. The value of this figure is that it makes recurring seasonal pressure visible. Instead of reading a long time-series line, the reader can see whether certain months repeatedly perform worse. This is especially useful for an industry like aviation, where demand peaks, school holidays, summer travel, and winter disruption can create recurring service stress.

The next group of results focuses on weather as an external attribution factor. Weather is important not only because it can disrupt operations, but because it changes how passengers assign responsibility.

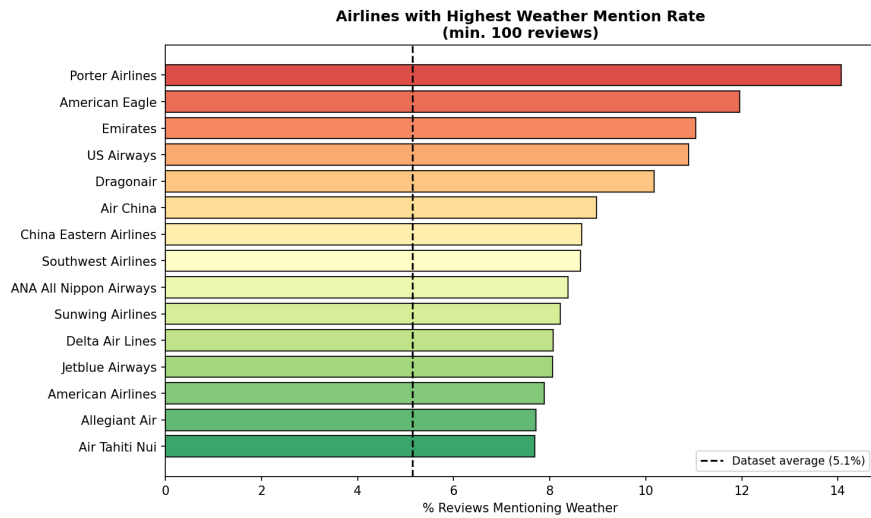


Figure 24: Airlines with the highest weather mention rate compared to the dataset average.

Figure 24 ranks airlines by weather mention rate and compares them with the dataset average. The figure should not be read simply as evidence that some airlines experience more bad weather. It is based on passenger mentions, not external meteorological records. Therefore, it shows where weather becomes part of the passenger narrative. This may reflect route geography, seasonality, operational exposure, or communication practices. The useful insight is that weather is not only an external condition; it is also part of how passengers explain the experience.

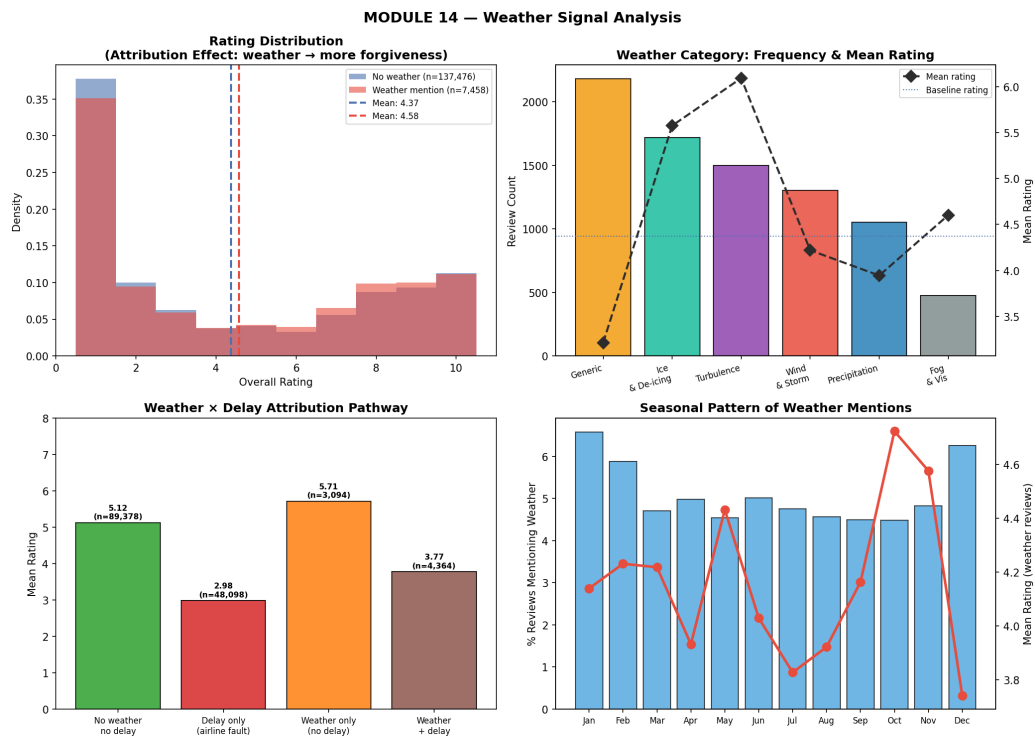


Figure 25: Weather signal analysis: rating distributions, attribution pathways, and seasonal mention patterns.

Figure 25 brings together several weather-related views. It shows how ratings differ for weather and non-weather reviews, how weather categories appear, and how mentions vary across time or season. The central story is that weather language can soften blame when passengers see disruption as externally caused. Weather-mentioning reviews score slightly higher on average than non-weather reviews. However, this does not mean weather improves satisfaction. Rather, it suggests that passengers may be more forgiving when the cause of a problem appears uncontrollable.

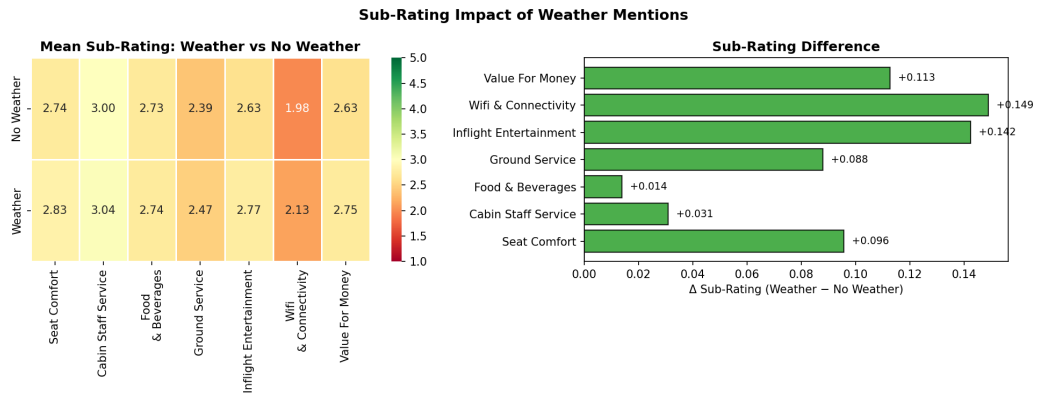


Figure 26: Impact of weather mentions across individual service sub-ratings.

Figure 26 shows how weather mentions relate to individual service sub-ratings. This figure is useful because it prevents the weather effect from being interpreted only through the overall rating. If weather changes passenger judgement, it may do so differently across service dimensions. The figure helps show whether weather affects operationally linked dimensions more than onboard service dimensions, and whether passengers separate uncontrollable disruption from controllable service recovery.

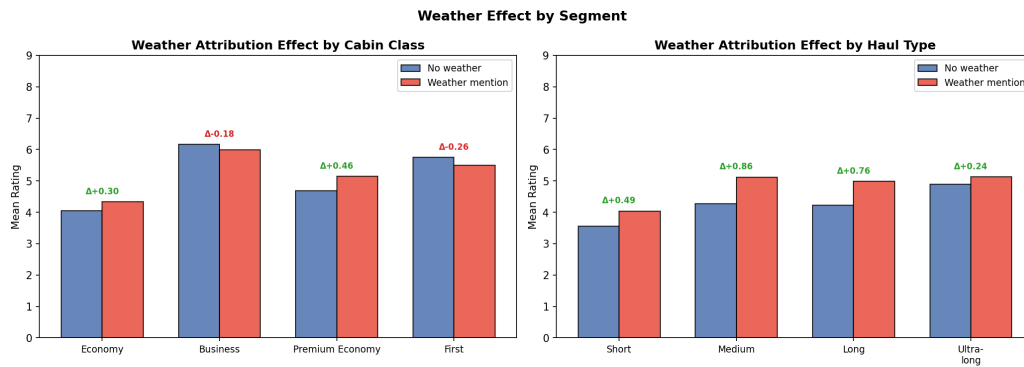


Figure 27: Weather attribution effect segmented by cabin class and haul type.

Figure 27 segments the weather effect by cabin class and haul type. This

is important because attribution may not operate equally across all passengers. Premium passengers may expect better communication and recovery even when weather is uncontrollable. Long-haul passengers may experience weather-related delays differently because the cost of disruption is higher. The figure therefore connects attribution to service expectations.

Table 12: Mean rating by weather–delay quadrant.

Condition	Mean rating	n
No weather, no delay	4.57	about 110,000
Delay only	2.64	about 28,000
Weather only	5.71	3,094
Weather + delay	3.77	4,364

Table [12](#) is one of the most important results for interpreting weather. It separates weather mentions from delay mentions, creating four conditions. Reviews with no weather and no delay have a mean rating of 4.57. Reviews with delay only fall sharply to 2.64. Reviews with weather only have the highest mean rating, 5.71, while reviews with both weather and delay fall to 3.77. The story is clear: weather alone is not damaging, and may even be associated with more forgiving evaluations. Delay is damaging. When weather and delay occur together, weather appears to soften the damage, but it does not remove it.

Table 13: Mean rating by weather category.

Category	n	Mean rating
Turbulence	1,499	6.09
Ice & De-icing	1,718	5.58
Fog & Visibility	474	4.60
Wind & Storm	1,301	4.22
Precipitation	1,052	3.94
Generic	2,181	3.21

Table 13 breaks the weather signal into categories. Turbulence has the highest mean rating, while generic weather language has the lowest. This gradient is meaningful. Turbulence is usually accepted as clearly uncontrollable, while vague expressions such as “weather delay” may appear when passengers are dissatisfied with the explanation or with the management of the disruption. The table therefore shows that not all weather language has the same meaning. Specific, clearly uncontrollable weather references are associated with more forgiveness; vague weather explanations are more likely to coexist with dissatisfaction.

The next result concerns peak season. Aviation is strongly seasonal, but global seasonality cannot be handled with a single calendar definition. The Northern Hemisphere and Southern Hemisphere have different summer peaks, so the analysis separates them.

Table 14: Peak vs off-peak satisfaction by hemisphere.

Hemisphere	Peak	Peak mean	Off-peak mean	Gap	p-value
Northern Hemisphere	Jun–Aug	3.54	4.01	-0.47	< 0.001
Southern Hemisphere	Dec–Feb	3.86	4.04	-0.18	0.094

Table 14 compares peak and off-peak ratings by hemisphere. The Northern Hemisphere shows a significant summer penalty of -0.47, while the Southern Hemisphere shows a smaller and not statistically significant gap. This table is useful because it quantifies what would otherwise remain a general intuition: high-demand travel periods are associated with lower satisfaction, especially in the Northern Hemisphere portion of the corpus.

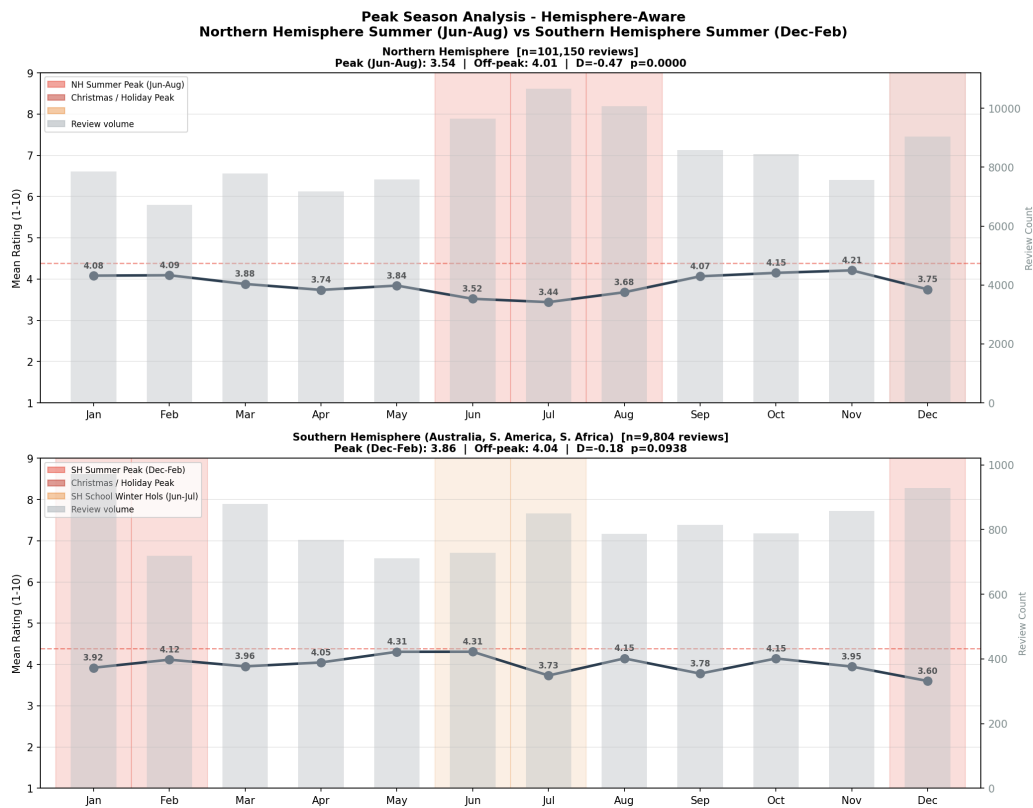


Figure 28: Hemisphere-aware peak season analysis: Northern vs. Southern Hemisphere review volume and mean ratings by month.

Figure 28 shows why the hemisphere-aware approach is necessary. The figure compares monthly patterns in the Northern and Southern Hemispheres. It should be read by looking at both review volume and mean rating across months. The Northern Hemisphere has far more observations and a clearer summer pattern. The Southern Hemisphere has fewer reviews, which reduces statistical power. The figure therefore explains both the substantive result and the limitation: the seasonal penalty is clear in the Northern Hemisphere, while the Southern Hemisphere pattern is directionally similar but less certain.

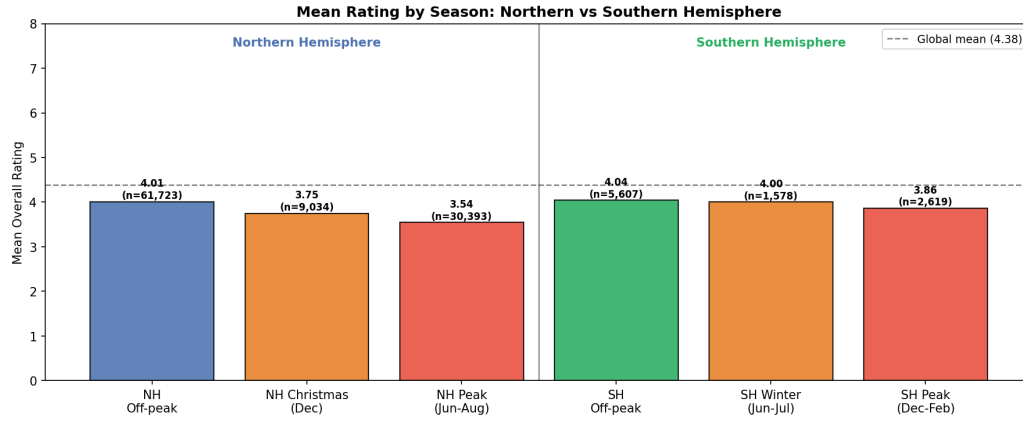


Figure 29: Mean overall rating by season across hemispheres.

Figure 29 summarises the seasonal comparison more directly. The bars show mean ratings by season and hemisphere. The useful information is the contrast between peak and off-peak periods. In the Northern Hemisphere, peak season clearly corresponds to lower satisfaction. For management, this suggests that service quality is vulnerable when demand is highest, which is exactly when many passengers are travelling for holidays, family visits, or other emotionally important trips.

The causal event analysis then examines whether major aviation events are associated with changes in satisfaction after adjusting for seasonal patterns.

Table 15: Difference-in-Differences estimates for seven aviation events.

Event	Pre	Post	Raw gap	DiD	Sig.
EU Aviation Strikes (2023-01)	2.55	3.24	+0.69	+1.329	***
Boeing 737 MAX Ban (2019-03)	3.71	3.43	-0.28	+0.803	***
Post-COVID Boom (2022-06)	2.45	2.70	+0.26	+0.730	***
CrowdStrike Outage (2024-07)	3.24	3.46	+0.22	+0.058	*
Boeing Door Blowout (2024-01)	3.20	3.52	+0.32	-0.301	***
Ukraine War & Airspace (2022-02)	2.69	2.52	-0.17	-0.345	n.s.
COVID-19 Pandemic (2020-03)	3.43	2.25	-1.17	-0.893	***

Table 15 reports the pre-event mean, post-event mean, raw change, Difference-in-Differences estimate, and significance for seven aviation events. The table should be read by comparing the raw gap with the DiD column. The raw gap shows the simple before-and-after movement in average ratings, while the DiD estimate asks a stronger question: whether that movement is different from the comparable seasonal movement in the previous year. This distinction is important because passenger satisfaction is not stable over the calendar year. Ratings naturally rise and fall with seasonality, demand pressure, and recovery periods, so a raw before-and-after comparison can easily give a misleading impression.

The Boeing Door Blowout provides the clearest example of why the Difference-in-Differences design is useful. In the raw before-and-after comparison, the mean rating appears to increase after the event, moving from 3.20 to 3.52. A simple interpretation would therefore suggest that satisfaction improved after the incident, which would not make substantive sense. However, the DiD estimate is negative, at -0.301 , because the event-window change is compared against the expected seasonal movement. In other words, ratings increased in absolute terms, but they increased less than expected once the seasonal baseline is taken into account. This is the value of DiD: it separates the apparent raw movement from the event-specific effect. The Boeing Door Blowout therefore illustrates how causal interpretation can change once the analysis controls for the background trend.

Other events help confirm the same logic. The COVID-19 Pandemic shows the strongest negative effect, with both a negative raw gap and a negative DiD estimate, which means that the decline is visible even before seasonal correction and remains substantial after correction. By contrast, the EU Aviation Strikes show a positive DiD estimate. This should not

be interpreted as strikes improving service quality. A more plausible interpretation is selection: passengers who still travelled or reviewed during the affected period may not represent the full population of disrupted passengers. Together, these examples show why the DiD table should not be read mechanically. The sign and size of each estimate must be interpreted together with the event context, the raw change, and the seasonal comparison.

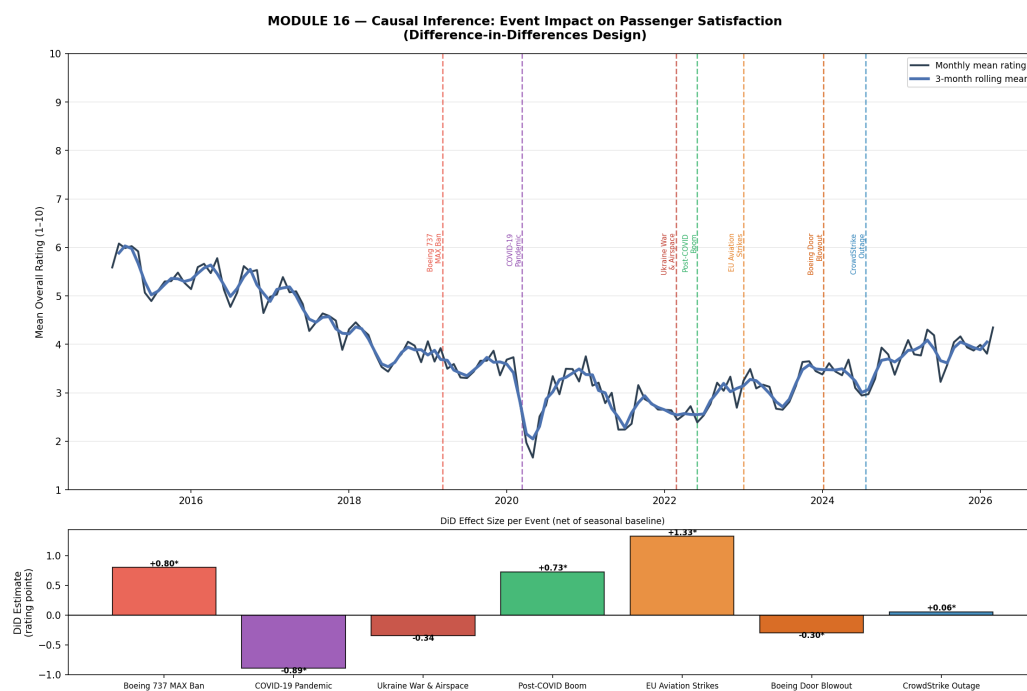


Figure 30: Causal inference mapping: event impact on passenger satisfaction using a Difference-in-Differences design.

Figure 30 visualises event impacts on the satisfaction timeline. The figure helps the reader see where events occur in relation to the broader movement of ratings. This matters because events do not happen in isolation. A disruption that occurs during a period of general recovery may look different from one that occurs during a period of decline. The figure therefore connects event-level interpretation with the long-run time-series context.

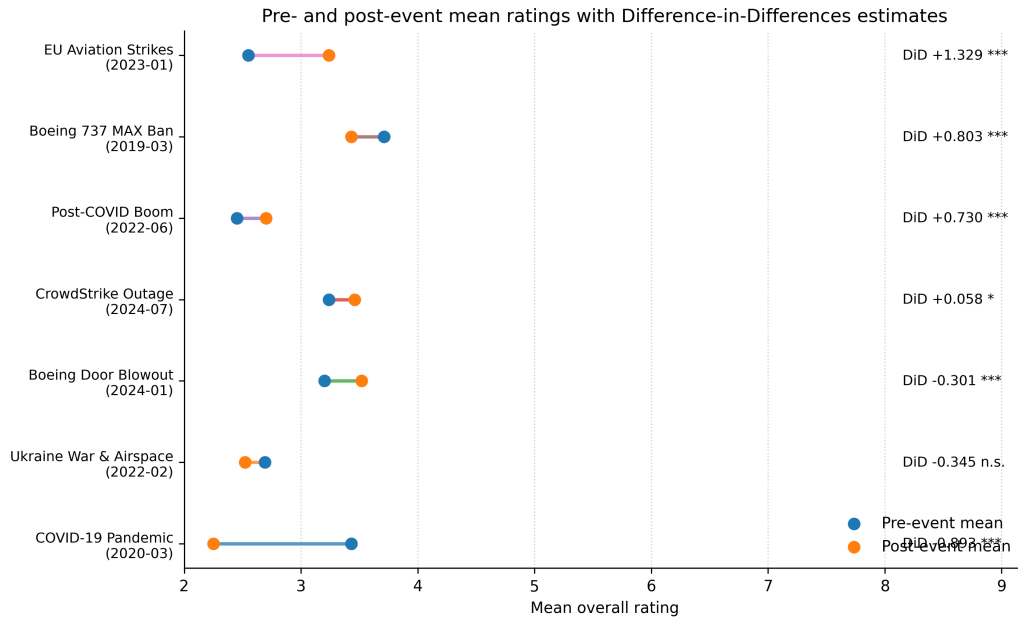


Figure 31: Pre- and post-event mean ratings across causal event windows.

Figure 31 presents the event-window results in a compact before-and-after format. Each comparison shows how the mean rating changes from the pre-event period to the post-event period, while Table 15 provides the corrected DiD estimate. This pairing is useful because it separates visual movement from causal interpretation. A raw increase or decrease is not enough on its own; it must be interpreted against the seasonal baseline.

The final result examines seasonal sensitivity by segment. Instead of asking only whether peak season reduces satisfaction, the analysis asks which passenger groups and journey types are most affected.

Table 16: Northern Hemisphere peak-season penalty by segment. Gap equals peak minus off-peak.

Dimension	Segment	Gap	Sig.
Carrier type	LCC	-0.42	***
Carrier type	Legacy	-0.50	***
Carrier type	Other	-0.39	***
Cabin class	Economy	-0.44	***
Cabin class	Premium Economy	-0.50	***
Cabin class	Business	-0.28	***
Cabin class	First	+0.05	n.s.
Haul type	Short-haul	-0.43	***
Haul type	Medium-haul	-0.45	***
Haul type	Long-haul	-0.39	***
Haul type	Ultra-long	-0.35	**

Table 16 shows the Northern Hemisphere peak-season penalty by carrier type, cabin class, and haul type. The table should be read as a set of gaps: negative values mean that peak-season ratings are lower than off-peak ratings. The penalty is widespread, but not equal across segments. Legacy carriers show a larger penalty than low-cost carriers, Premium Economy shows the largest cabin-class penalty, and First Class shows no significant decline. This suggests that the peak-season problem is not simply an operational capacity issue. It is also an expectations issue. Passengers who pay for a service differential may react strongly when that differential narrows during periods of high demand.

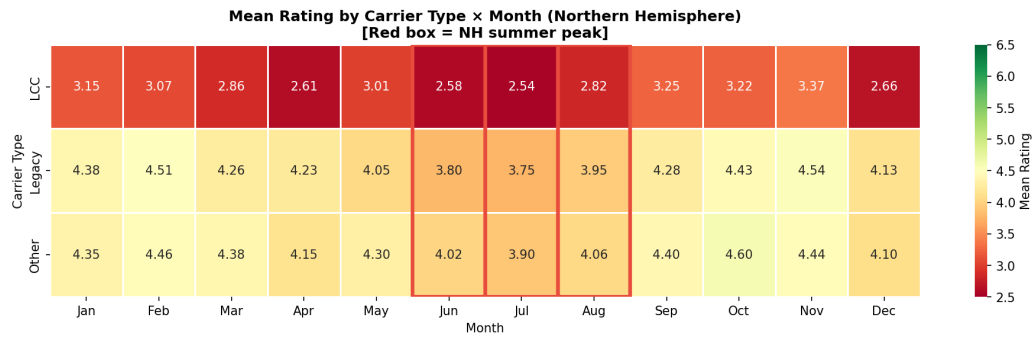


Figure 32: Mean rating by carrier type and month in the Northern Hemisphere.

Figure 32 shows mean ratings by carrier type and month. Like the earlier seasonality heatmap, it should be read by comparing colour intensity across months and carrier groups. The figure is useful because it shows whether seasonal deterioration is concentrated in specific carrier types or spread across the market. It reinforces the idea that seasonality is not only a calendar effect; it interacts with business model and passenger expectation.

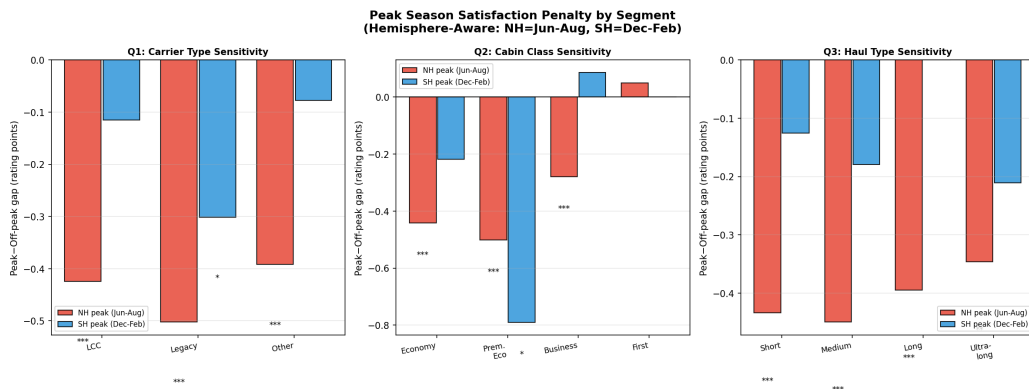


Figure 33: Peak season satisfaction penalty segmented by carrier type, cabin class, and haul type.

Figure 33 translates the seasonal gaps into a visual comparison across segments. The negative bars show where ratings fall during peak season. This figure is especially useful for management because it identifies where

seasonal stress is most damaging. Premium Economy stands out because passengers pay more than Economy but may not receive enough protection from service deterioration during peak periods. Business Class is affected less, and First Class appears insulated. This suggests that service floors matter: the stronger and more protected the service product, the less vulnerable it is to seasonal degradation.

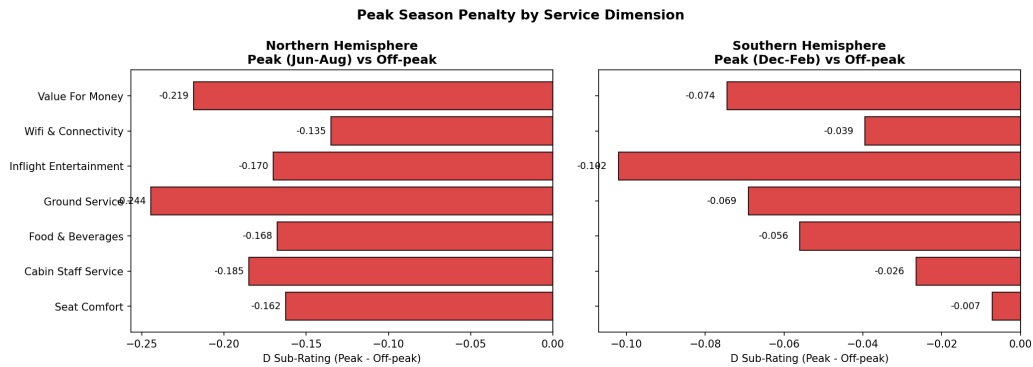


Figure 34: Peak season penalty by individual service dimension.

Figure 34 shows the peak-season penalty by service dimension. This figure is useful because it moves from passenger segments to operational aspects. It asks which parts of the service experience deteriorate most during peak demand. For airlines, this type of figure is directly actionable. If the peak-season penalty is concentrated in Ground Service, operational staffing and airport communication may be the priority. If it appears in Value For Money, the issue may be expectation management and fare perception. If it affects Cabin Comfort or onboard service, then the pressure may be related to load factors, cabin density, or reduced service attention.

Taken together, the results show that passenger satisfaction is shaped by several layers of evidence. The core ABSA results confirm that airline reviews are overwhelmingly multi-aspect and that TF-IDF with Logistic Regression provides a stronger and more robust classifier than both a simple lexicon

baseline and a fine-tuned DistilBERT model under silver-label supervision. The rating-alignment and predictive results identify Value For Money as the dominant satisfaction driver. The AOPE results show that Punctuality is the only aspect with clearly negative language, making it a non-compensable damage factor. The contextual analyses then show that satisfaction is not stable across business model, cabin class, route length, aircraft type, delay, traveller segment, weather attribution, seasonality, or global events.

The chapter therefore moves from model evaluation to managerial interpretation. The technical results answer whether the ABSA pipeline works; the contextual results explain what the pipeline reveals about the passenger experience. The overall story is that airline satisfaction cannot be understood from a single rating or a single sentiment score. It is a multidimensional judgement shaped by service delivery, perceived fairness, operational reliability, external attribution, and demand pressure.

Chapter 7

Discussion

This chapter returns to the three research questions introduced at the beginning of the thesis and interprets the empirical results presented in Chapter 6. The purpose is not to repeat every figure and table, but to connect the evidence into a coherent answer to the research problem. Chapter 6 showed that the proposed pipeline is able to move from raw passenger reviews to aspect-level sentiment, rating alignment, contextual segmentation, and event-based interpretation. The discussion now asks what these results mean for Aspect-Based Sentiment Analysis in the airline domain and for the managerial understanding of passenger satisfaction.

7.1 RQ1: Domain-Aware ABSA and the Transformer Comparison

The first research question asked how domain-aware ABSA performs compared to a lexicon baseline, and whether fine-tuned DistilBERT improves over TF-IDF on domain-specific airline text. The results show that domain-aware ABSA provides a clear improvement over a generic lexicon approach, but they do not support the expectation that the transformer model automatically improves over the classical TF-IDF classifier.

The first part of the answer comes from the aspect extraction stage. Table 4 and Figure 5 show that the BIO tagger identified 717,904 aspect spans across 138,813 reviews, corresponding to 95.8% review coverage. This is an important result because it confirms that the selected aspect schema captures the main service dimensions passengers discuss in airline reviews. The fact that 76.2% of reviews contain two or more distinct aspects also supports the basic motivation of the thesis: airline reviews cannot be adequately represented by a single document-level sentiment label. They are multi-aspect narratives in which passengers may evaluate punctuality, crew, comfort, food, and value differently within the same text.

The second part of the answer comes from the comparison between the lexicon baseline and the TF-IDF model. Table 5 shows that the lexicon baseline reaches a macro-F1 of 0.462, while TF-IDF with Logistic Regression reaches an overall macro-F1 of 0.608. Figure 6 makes this improvement visible across the five aspects. This means that the domain-aware supervised model captures patterns that are not available to a simple sentiment dictionary. Airline reviews often express dissatisfaction through events rather than through isolated sentiment words. A phrase describing a missed connection, a long wait on the tarmac, or poor boarding communication may be negative even if it does not contain a standard negative adjective. The improvement over the lexicon baseline therefore confirms that airline ABSA benefits from learning domain-specific textual associations.

However, the transformer comparison complicates the interpretation. Table 6 and Table 7 show that fine-tuned DistilBERT does not outperform TF-IDF overall. DistilBERT reaches an overall macro-F1 of 0.574, compared with 0.608 for TF-IDF. Figure 8 shows the same pattern visually: DistilBERT is close to TF-IDF on several aspects, and approximately equal on

Food & Beverages, but it does not produce a general improvement. The result is particularly important because it goes against the common assumption that a transformer model will necessarily dominate a classical feature-based model.

The most plausible explanation is that the bottleneck is not model capacity, but label quality. The sentiment labels used in this thesis are silver labels derived from structured ratings. This makes large-scale training possible, but it also introduces noise because a sub-rating does not always perfectly correspond to the local sentiment expressed in the text. A passenger may give a low overall score because of a delay, while still writing positively about the crew. A high-capacity model such as DistilBERT may be more exposed to this inconsistency, while TF-IDF with Logistic Regression is simpler, more regularised, and more robust under noisy supervision.

The confusion matrices in Figure 9 provide further evidence for this interpretation. DistilBERT performs better on clearly negative and clearly positive cases than on neutral cases. The neutral class is the main bottleneck, especially for aspects such as Punctuality and Value For Money. This is not surprising in the airline domain. Neutral sentiment is rarely a clear linguistic category. It often represents mixed experience, mild dissatisfaction, or compensation between positive and negative service elements. A transformer can model context, but it cannot fully solve ambiguity when the training labels themselves are indirect.

The disagreement analysis in Table 8 strengthens this conclusion. The highest lexicon–TF-IDF disagreement occurs for Value For Money, where the disagreement rate reaches 59.8%. This confirms that some aspects are not easily interpreted through surface sentiment words. Value judgements depend on price, expectations, cabin class, route, and perceived fairness. By

contrast, Cabin Crew Service has the lowest disagreement rate, suggesting that human service is often described with more direct evaluative vocabulary. Overall, RQ1 is answered as follows: domain-aware ABSA substantially improves over a lexicon baseline, but DistilBERT does not improve over TF-IDF under the silver-label conditions of this study. The result supports the importance of domain adaptation, but also shows that better model architecture cannot fully compensate for noisy or indirect labels.

7.2 RQ2: Satisfaction Drivers and Segment Variation

The second research question asked which airline service aspects most strongly influence overall customer satisfaction, and how sentiment distributions vary across aspects, carrier types, cabin classes, and traveller segments. The results point to a clear hierarchy of satisfaction drivers. Value For Money is the strongest global driver of satisfaction, while Punctuality operates as the most damaging failure dimension. Segment-level results then show that satisfaction is shaped by the service promise associated with carrier type, cabin class, traveller type, route, and aircraft category.

The strongest evidence for Value For Money comes from three independent parts of the analysis. First, Table 9 shows that Value For Money has the highest Pearson correlation with the overall rating, with $r = 0.899$. Figure 10 confirms this visually by placing Value For Money at the top of the correlation ranking. Second, Figure 11 shows that Value For Money is also the dominant predictor in the Ridge regression model, which achieves $R^2 = 0.874$ on the held-out post-2023 test set. Third, Table 11 shows that Value For Money has a near-neutral AOPE polarity of +0.17, making it the most ambivalent aspect in the language-level analysis.

Taken together, these results show that Value For Money is not simply another service attribute. It is the interpretive lens through which passengers judge the entire exchange. Passengers do not evaluate comfort, food, crew, or punctuality in isolation. They interpret them relative to the fare paid, the cabin class selected, the carrier type, and the service promise they expected. This explains why Value For Money can be both the strongest statistical driver and the most linguistically ambiguous aspect. It is powerful because it summarises the perceived fairness of the whole journey, but it is difficult to classify because fairness depends on context.

Punctuality plays a different role. Table 11 shows that Punctuality is the only aspect with negative mean AOPE polarity, at -1.03 . Figure 20 shows that “delayed” is the most frequent opinion word associated with the extracted aspects. Figure 15 further shows that delay behaves like a satisfaction switch: ratings fall sharply once delay reaches a serious threshold and then remain low. This suggests that Punctuality is a non-compensable damage factor. A good meal, friendly crew, or comfortable seat may improve the journey, but they may not fully repair the dissatisfaction created by a severe delay or missed connection.

The segment analyses show that satisfaction also depends on the context in which the passenger evaluates the airline product. Figure 12 shows that legacy carriers perform slightly better than low-cost carriers on several service dimensions, but that Value For Money remains very close between the two groups. This implies that passengers do not judge all airlines against the same standard. Low-cost passengers may accept a more limited product if the fare is perceived as fair, while legacy passengers expect a stronger service offer because the brand promise and price level are different.

Cabin class produces a similar expectation effect. Table 10, Figure 13, and

Figure 14 show that Business Class generally performs better than Economy across service dimensions. However, the difference is not uniform across all aspects. Wifi & Connectivity remains weak across all cabins, and Economy Wifi & Connectivity is the lowest sub-rating in the cabin-class table. This indicates that some weaknesses are structural rather than confined to a single cabin class. The premium cabins improve the service level, but they do not eliminate all dissatisfaction, especially when passengers evaluate dimensions such as connectivity against modern expectations.

Traveller type adds another layer. Figure 17 shows that satisfaction varies when traveller type and cabin class are considered together. Solo Business travellers in Business Class record the highest satisfaction, while Family Leisure passengers in Economy record the lowest. This reflects different needs and different exposure to friction. Business travellers may value efficiency and reliability, while families may face more stress around seating, boarding, delays, and service support. Figure 18 shows that when delay is mentioned, these differences narrow because all traveller groups converge toward lower satisfaction. This means that delay reduces the importance of baseline segment expectations and creates a shared dissatisfaction floor.

The route and aircraft results further support the idea that satisfaction is contextual. Figure 16 shows that short-haul routes have lower satisfaction than medium-, long-, and ultra-long-haul routes, while the positive sentiment share increases with distance. Figure 19 shows that widebody aircraft perform better than narrowbody aircraft, especially on dimensions linked to the onboard product such as entertainment. These results suggest that the physical and operational structure of the journey shapes satisfaction alongside the airline brand itself. Overall, RQ2 is answered as follows: Value For Money is the strongest overall satisfaction driver, Punctuality is the clearest

negative damage factor, and satisfaction varies systematically across carrier type, cabin class, traveller type, route length, and aircraft category because passengers evaluate service relative to expectations and journey context.

7.3 RQ3: Contextual Modulation of Satisfaction

The third research question asked to what extent contextual factors—weather, seasonal demand, and global disruption events—modulate the relationship between aspect-level sentiment and overall satisfaction. The results show that contextual factors do not merely add background variation. They actively reshape how passengers interpret service outcomes, especially when disruption, attribution, demand pressure, and external shocks are involved.

Weather provides the clearest example of contextual reinterpretation. Figure 25 shows that weather-related language is associated with a slightly higher average rating than non-weather language, but this does not mean that weather improves satisfaction. Rather, it suggests that passengers may assign less blame to the airline when the cause of disruption appears external and uncontrollable. Table 12 clarifies the mechanism. Reviews with no weather and no delay have a mean rating of 4.57, while delay-only reviews fall to 2.64. Weather-only reviews have the highest mean rating, 5.71, while weather plus delay reviews fall to 3.77. The important point is that delay remains damaging, but weather appears to soften the damage when passengers interpret the disruption as externally caused.

Table 13 shows that not all weather language has the same meaning. Turbulence has the highest mean rating, while generic weather references have the lowest. This suggests that specific and clearly uncontrollable weather conditions may trigger more passenger forgiveness, whereas vague weather

explanations may coexist with dissatisfaction or distrust. Figure 26 and Figure 27 further show that weather attribution interacts with service dimensions, cabin class, and haul type. Premium passengers and long-haul passengers may accept uncontrollable disruption, but they may still expect stronger communication and recovery. Weather therefore modulates satisfaction through attribution: it changes how passengers assign responsibility, not simply what happened operationally.

Seasonality provides a second form of contextual modulation. Table 14, Figure 28, and Figure 29 show that the Northern Hemisphere peak season produces a significant summer penalty. The peak mean is 3.54, compared with an off-peak mean of 4.01, giving a gap of -0.47 with $p < 0.001$. The Southern Hemisphere shows a smaller and not statistically significant gap. This result shows that satisfaction is lower when demand pressure is highest, especially in the portion of the corpus where summer travel volume is largest. The seasonal effect is not only a calendar pattern. It reflects the interaction between high passenger volumes, constrained operational capacity, airport congestion, fare expectations, and emotionally important travel periods.

The segment-level seasonal results deepen this interpretation. Table 16, Figure 32, Figure 33, and Figure 34 show that the peak-season penalty is widespread but uneven. Legacy carriers show a larger peak-season penalty than low-cost carriers, Premium Economy shows the largest cabin-class penalty, and First Class shows no significant decline. This pattern suggests that peak season does not affect all passengers equally. It damages satisfaction most where expectations are high enough to create disappointment but the service floor is not strong enough to protect the passenger experience. Premium Economy is especially exposed because passengers pay for a service differential that may become less visible under peak-season stress.

Global disruption events provide the third contextual layer. Table 15, Figure 30, and Figure 31 show that major aviation events can shift satisfaction, but that raw before-and-after comparisons can be misleading. The Boeing Door Blowout is the clearest example. The raw mean rating increases after the event, but the Difference-in-Differences estimate is negative. This means that ratings increased less than expected after accounting for the seasonal baseline. The example shows why DiD is necessary: it separates apparent raw movement from event-specific impact.

The COVID-19 Pandemic shows the strongest negative event effect, which is consistent with the scale of the disruption it created for airline operations and passenger confidence. By contrast, the EU Aviation Strikes show a positive DiD estimate. This should not be interpreted as strikes improving service quality. A more plausible explanation is selection: the passengers who continued to travel or who wrote reviews during the strike window may not represent the full population of disrupted passengers. This interpretation connects the event analysis back to the self-selection dynamics of online reviews discussed in Chapter 4. Overall, RQ3 is answered as follows: contextual factors substantially modulate the relationship between aspect-level sentiment and satisfaction. Weather changes attribution, seasonality changes service pressure and expectations, and global events change the baseline against which passengers evaluate their journeys.

7.4 Cross-Module Synthesis

Across the three research questions, the thesis shows that airline passenger satisfaction is best understood as a layered judgement rather than a single sentiment outcome. The first layer is textual and aspect-based. The BIO

extraction results show that airline reviews contain multiple service dimensions, and the ASC results show that domain-aware modelling improves over a generic lexicon baseline. The second layer is evaluative. Correlation and Ridge regression show that Value For Money dominates overall satisfaction, while AOPE shows that Punctuality carries the strongest negative local language. The third layer is contextual. Carrier type, cabin class, traveller type, route, aircraft, weather, seasonality, and disruption events all influence how passengers interpret the same service outcome.

The most important conceptual result is the distinction between compensable and non-compensable dimensions. Value For Money is compensable in the sense that passengers evaluate it through trade-offs. A limited service may still be acceptable if the fare is low, while a better service may still be judged poorly if the price feels unfair. Punctuality is different. Severe delay appears to create a dissatisfaction floor that other aspects struggle to offset. This explains why Punctuality has negative AOPE polarity and why delay produces convergence across traveller groups. It also helps explain why satisfaction cannot be managed only through cabin product improvements. Operational reliability remains central.

A second synthesis concerns model choice. The thesis does not show that classical models are always better than transformers. Rather, it shows that in a large real-world corpus with silver labels, a simpler and more regularised model can be more reliable than a more complex neural model. This has practical implications for organisations using customer-review data. Before investing in larger models, it may be more valuable to improve label quality, validate aspect-level annotations, and understand the ambiguity of the target categories. In this study, the transformer comparison becomes evidence for a broader methodological point: model complexity should match the quality

and structure of the supervision.

A third synthesis concerns managerial relevance. The later modules of the pipeline demonstrate that ABSA becomes more useful when it is connected to operational and contextual variables. The value of the analysis is not only that it classifies sentiment toward Cabin Comfort or Food & Beverages. Its value lies in showing when and for whom those sentiments matter most. Low-cost and legacy passengers apply different expectations. Economy and Business passengers experience different service floors. Families and business travellers react differently until delay creates a shared dissatisfaction floor. Weather can soften blame, while peak season can intensify disappointment. These findings turn passenger reviews into structured managerial evidence.

The overall contribution of the thesis is therefore twofold. Methodologically, it demonstrates that a domain-aware, modular ABSA pipeline can extract meaningful aspect-level and contextual evidence from large-scale airline reviews. Substantively, it shows that airline satisfaction is governed by a combination of perceived fairness, operational reliability, expectation management, and contextual attribution. The answers to the research questions are therefore connected: the ABSA pipeline works because airline reviews are multi-aspect; the most important drivers are Value For Money and Punctuality because passengers evaluate both fairness and reliability; and contextual factors matter because passengers do not judge service outcomes independently of cause, timing, demand pressure, or disruption history.

Chapter 8

Conclusions, Limitations, and Future Work

This final chapter brings the thesis to a close by returning to the central idea that motivated the entire study: airline passenger satisfaction cannot be understood through a single overall sentiment score. Air travel is not experienced as one undifferentiated service, but as a sequence of expectations, interactions, operational events, physical conditions, emotional reactions, and judgements of fairness. For this reason, the passenger’s final evaluation is built across several dimensions at once. A review may describe a delayed departure, a helpful crew, an uncomfortable seat, and a disappointing sense of value within the same narrative. The purpose of this thesis has been to show that these different layers can be analysed systematically through a domain-aware Aspect-Based Sentiment Analysis framework.

The aim of this chapter is therefore not to repeat all the empirical outputs of Chapter 6 or the research-question synthesis of Chapter 7, but to step back from the individual modules and explain the broader meaning of the work. It reflects on the main contribution of the study, discusses the limitations that shape the interpretation of the findings, identifies directions for future research, and closes with the managerial implications of the results.

The project proposed an end-to-end framework capable of moving from raw passenger reviews to structured evidence about airline service quality. The pipeline extracted aspect mentions, classified their sentiment, compared classical and neural models, aligned textual signals with numerical ratings, and extended the analysis to carrier type, cabin class, route length, aircraft type, traveller segment, weather attribution, peak-season pressure, and major aviation events. In this sense, the contribution of the thesis is both methodological and managerial. Methodologically, it shows how Aspect-Based Sentiment Analysis can be adapted to a complex service domain. Managerially, it shows how passenger reviews can reveal which parts of the experience matter most, when dissatisfaction becomes most severe, and under which conditions passengers reinterpret service failures.

A central lesson of the thesis is that passenger reviews are neither pure noise nor perfect measurement. They are imperfect but valuable traces of lived service experiences. They are subjective, self-selected, emotional, and sometimes incomplete, but precisely for these reasons they contain information that structured indicators often miss. Operational data can show whether a flight departed late, but a review can show whether the passenger felt abandoned, misinformed, treated fairly, or ignored. A sub-rating can show that a passenger judged Ground Service poorly, but the text can explain whether the issue was boarding, queuing, gate staff, baggage, re-booking, or communication. The value of the thesis lies in connecting these layers: the numerical rating, the aspect-level signal, the written narrative, and the context in which the journey occurred.

8.1 Overall Contribution and Principal Conclusions

The first major conclusion is that domain-aware ABSA is necessary for airline reviews. The results in Chapter 6 showed that reviews are overwhelmingly multi-aspect. The BIO tagger identified 717,904 aspect spans across 138,813 reviews, with 95.8% review coverage, and Figure 5 showed that most reviews mention more than one service dimension. This confirms the central premise of the thesis. Airline passengers rarely evaluate only one part of the journey. They may describe a delay, then praise the crew, then criticise the seat, and finally judge the ticket price as unfair. A document-level sentiment model would collapse these different evaluations into one label, losing precisely the information that is most useful for service improvement.

This conclusion is also consistent with the theoretical argument developed in the early chapters. Airline satisfaction is not formed at a single moment. It develops through the journey, beginning with booking and price expectations, continuing through airport processes and onboard service, and ending with arrival, baggage, and the passenger’s retrospective interpretation of the experience. A passenger review is therefore not simply an opinion about an airline brand. It is a compact narrative of a service chain. This is why aspect-level analysis is so important. It allows the review to be decomposed into service dimensions without losing the fact that these dimensions interact within one passenger experience.

The model comparison provides the second major conclusion. The TF-IDF and Logistic Regression classifier clearly outperformed the lexicon baseline, as shown in Table 5 and Figure 6. This means that airline-specific language contains patterns that cannot be captured adequately by a generic dictionary of positive and negative words. In airline reviews, dissatisfaction

is often expressed through events rather than adjectives. A passenger who writes that they missed a connection, waited for hours, or received no information at the gate may express strong dissatisfaction without using many conventional sentiment words. A domain-aware supervised model is therefore better suited to the structure of the data.

At the same time, the comparison with DistilBERT produced an important methodological lesson. As shown in Table 6, Table 7, and Figure 8, the fine-tuned transformer did not outperform the simpler TF-IDF model overall. This result should be interpreted carefully. It does not mean that transformers are unsuitable for ABSA, nor that DistilBERT is a weak model. Rather, it shows that model capacity is not always the binding constraint. In this thesis, the labels used for aspect sentiment classification were silver labels derived from structured ratings. They made large-scale modelling possible, but they were not manually annotated aspect-level judgements. Under this type of supervision, a higher-capacity model can learn inconsistencies in the labels rather than more meaningful linguistic patterns. The result therefore warns against assuming that more complex models automatically produce better applied insight. In real organisational data, label quality, task definition, and domain fit may matter as much as architecture.

This finding is particularly relevant for organisations adopting artificial intelligence in customer-experience analytics. There is often a temptation to treat model complexity as a substitute for careful problem formulation. The evidence in this thesis suggests the opposite. The most powerful model is not necessarily the most useful model if the supervision is noisy, if the target construct is ambiguous, or if the output is difficult to interpret. In the airline review setting, a transparent TF-IDF model trained on domain-specific evidence produced stronger and more stable results than a contextual

transformer trained on the same silver labels. This does not close the door to neural modelling; rather, it clarifies that future improvements should begin with better labels and clearer annotation design before moving to larger architectures.

The third major conclusion concerns satisfaction drivers. Passenger satisfaction is dominated by perceived value and damaged most strongly by operational failure. Value For Money emerged as the strongest satisfaction driver across multiple parts of the analysis. It had the highest correlation with the overall rating in Table 9, it was the dominant predictor in the Ridge regression shown in Figure 11, and it was the most ambivalent aspect in the AOPE polarity table. This consistency matters because it shows that Value For Money is not merely a financial variable. It is a judgement about fairness. Passengers ask whether the service delivered was acceptable relative to the price paid, the cabin class selected, the brand promise of the airline, and the expectations created before the journey. For managers, this means that satisfaction cannot be improved only by adding service features. Passengers must also feel that the exchange was fair.

Punctuality plays a different role. It does not behave like a normal service attribute that can easily be balanced against other positives. Table 11 showed that Punctuality was the only aspect with negative mean AOPE polarity, and Figure 20 showed the dominance of delay-related vocabulary in passenger language. Figure 15 then showed that delay acts as a sentiment switch: once disruption becomes serious enough, ratings fall sharply and remain low. This suggests that punctuality is a non-compensable damage factor. A friendly crew or a comfortable seat may soften a negative journey, but they do not fully erase the effect of a severe delay, a missed connection, or a lack of information during disruption.

The distinction between Value For Money and Punctuality is one of the most useful interpretive outcomes of the thesis. Value For Money is a global judgement of exchange. It is shaped by trade-offs, expectations, and comparison. A limited service can still be acceptable if the fare is low and expectations are clear. A premium service can still feel disappointing if the passenger believes the price was not justified. Punctuality, by contrast, is less negotiable. It is closely tied to trust, reliability, and control. When punctuality fails, the passenger may experience not only inconvenience, but loss of time, missed commitments, anxiety, and reduced control over the journey. This helps explain why punctuality appears so strongly in negative language and why delays can compress satisfaction differences across traveller groups.

The fourth major conclusion is that satisfaction is contextual. Passengers do not evaluate every airline, cabin, route, or disruption according to the same standard. The LCC versus legacy comparison in Figure 12 showed that Value For Money remains close across business models even when other service dimensions differ. This suggests that passengers adapt their expectations to the carrier type. A low-cost passenger may accept a more limited product if the price feels fair, while a legacy passenger may judge the same service more severely because the promise is different. Cabin class results in Table 10, Figure 13, and Figure 14 similarly showed that higher cabins perform better, but do not eliminate dissatisfaction. Even premium passengers may become dissatisfied when the experience fails to meet the higher promise attached to the fare.

Traveller type, route length, and aircraft type add further nuance. Figure 17 showed that satisfaction depends on the interaction between traveller type and cabin class, while Figure 18 showed that delay reduces those differences by pushing all groups toward lower satisfaction. Figure 16 and

Figure 19 showed that the physical and operational context of the journey also matters. Short-haul flights, narrowbody aircraft, limited onboard services, and high-density operating models create different expectations and constraints from long-haul or widebody services. The passenger experience is therefore not only a matter of airline brand; it is also shaped by the service environment in which the journey takes place.

The fifth major conclusion is that external context changes how passengers assign blame. Weather results were especially important because they showed that disruption is not interpreted only through its outcome, but also through its perceived cause. Table 12 showed that delay-only reviews have very low mean ratings, while weather-only reviews have higher ratings and weather-plus-delay reviews sit between these two extremes. Table 13 added that specific weather categories are interpreted differently, with clearly uncontrollable references such as turbulence receiving higher ratings than generic weather language. This supports the idea that attribution matters: passengers may tolerate disruption more when they believe the cause is external, but they still judge the airline on communication, recovery, and care.

Seasonality and aviation events also changed the interpretation of satisfaction. Table 14, Figure 28, and Figure 29 showed that Northern Hemisphere peak season is associated with a significant satisfaction penalty. Table 16 and Figure 33 showed that this penalty is not evenly distributed: Premium Economy and legacy carriers are especially exposed, while First Class appears more insulated. The Difference-in-Differences analysis in Table 15 and Figure 31 showed that major events can also shift satisfaction in ways that are not visible from raw before-and-after comparisons. The Boeing Door Blowout

is the clearest example: the raw change appears positive, but the DiD estimate is negative once the seasonal baseline is considered. This demonstrates why contextual and causal thinking are necessary when interpreting review data.

Taken together, the thesis shows that airline satisfaction is a layered judgement. It is built from aspect-level service experiences, filtered through value expectations, damaged by operational failure, softened or intensified by attribution, and reshaped by demand pressure and external shocks. This is the central message of the research. Passenger reviews are not just complaints or compliments. When analysed systematically, they become evidence about how passengers understand the airline product.

8.2 Limitations and Future Research Directions

The results of this thesis should be interpreted in light of several limitations. These limitations do not invalidate the findings, but they define the boundaries of what can be claimed. They also indicate the most promising directions for future work. Because the study is based on large-scale observational review data, the strongest conclusions concern patterns, associations, comparative performance, and plausible interpretations. The analysis provides rich evidence about passenger perception, but it should not be confused with a controlled experiment or a fully representative passenger survey.

The first limitation concerns the nature of the data. The corpus is based on public online reviews, and public reviews are not equivalent to a random sample of all passengers. As discussed in Chapter 4, voluntary review platforms tend to overrepresent intense experiences. Passengers are more likely to write when they are very dissatisfied or very satisfied, while routine flights

often remain invisible. This means that the corpus should be interpreted as a record of experiences that were strong enough to motivate public feedback, not as a direct measure of the average satisfaction of all airline passengers. This limitation is especially important for negative findings. Delay, poor communication, and service failures may appear particularly severe because passengers affected by these problems are also more likely to write reviews.

This limitation is not only statistical, but behavioural. The decision to write a review is part of the customer experience itself. A passenger may publish a review because they want compensation, because they want to warn others, because they feel ignored by the airline, or because they want to reward exceptional service. The review therefore reflects both the journey and the motivation to speak publicly about it. This does not make the data unusable. On the contrary, it makes the data valuable for understanding salient experiences. However, it means that the results should be interpreted as evidence about visible and memorable passenger experiences rather than as a perfectly balanced record of all flights.

A related limitation concerns market and language coverage. The corpus is based on reviews available through a specific platform and is predominantly English-language. This may overrepresent passengers from English-speaking markets or passengers comfortable writing in English. It may also underrepresent experiences from regions where passengers use different review platforms or different languages. Airline satisfaction is culturally and geographically variable, and expectations around service, compensation, punctuality, and communication may differ across markets. Future work could therefore extend the framework to multilingual corpora and additional platforms. This would make it possible to test whether the findings generalise beyond the English-language review environment.

The second major limitation concerns silver label quality. This is the most important methodological limitation of the modelling stage. The aspect sentiment labels were derived from structured ratings rather than from manual annotation. This allowed the pipeline to scale to a large corpus, but it also introduced ambiguity. A sub-rating is an imperfect proxy for local textual sentiment. For example, a passenger may give a low Ground Service score because of boarding confusion, while the review text also praises the staff member who eventually solved the problem. In such cases, the numerical rating and the local sentiment may not align perfectly. The DistilBERT result provides indirect evidence of this limitation. If the labels were clean and conceptually precise, the transformer might have been expected to outperform the classical model. Instead, TF-IDF proved more robust.

Future work should therefore begin with manual annotation of a gold-standard subset of reviews. Even a relatively small but carefully annotated dataset would be valuable. It would allow the performance of the BIO extractor and the sentiment classifiers to be evaluated against true aspect-level labels rather than rating-derived proxies. It would also clarify whether DistilBERT underperformed because of label noise, task ambiguity, model configuration, or insufficient context. A manually annotated subset could then be used to fine-tune or calibrate larger models, validate neutral-class definitions, and measure the true error structure of the pipeline.

The neutral class deserves particular attention in future research. The confusion matrices showed that neutral sentiment is the most difficult category, especially for DistilBERT. This is partly because neutrality in airline reviews is often not true neutrality. It may represent mixed experience, weak dissatisfaction, compensation between different service elements, or a moderate rating produced by conflicting emotions. Future annotation guidelines

should therefore distinguish between genuinely neutral language, mixed sentiment, and low-intensity positive or negative sentiment. This would make the ASC task conceptually cleaner and could improve both classical and neural models.

The third limitation concerns aspect extraction. The BIO tagger used in this thesis is transparent, reproducible, and domain-aware, but it is still based on predefined triggers. This makes it interpretable, which is useful in an applied thesis, but it may miss paraphrastic or implicit references. A passenger may describe punctuality without using explicit delay terms, or refer to comfort through expressions such as “no space for my legs” rather than through predefined keywords. These cases are difficult for a rule-based extractor. Future work could therefore test neural sequence-labelling models, such as a BERT-CRF architecture, to improve recall on implicit or paraphrastic aspect mentions. The trade-off would be interpretability: a neural extractor may find more mentions, but its decisions may be harder to explain than a trigger-based system.

A further development would be to expand the aspect schema itself. The thesis focused on five extracted textual aspects and seven structured sub-rating dimensions. This structure was appropriate for the available data and for the core research questions, but airline reviews contain additional themes that could be modelled more explicitly. Baggage handling, customer service after disruption, booking systems, loyalty programmes, boarding organisation, airport transfer experience, and communication quality could each become distinct aspects in a more granular schema. Communication, in particular, appears repeatedly across several results. It matters during delays, weather disruption, peak-season stress, and event shocks. Future work could therefore treat communication not merely as a supporting explanation, but

as a separate aspect of passenger satisfaction.

The fourth limitation concerns the anomalous review detection module. The analysis can identify reviews that are unusual according to content signals such as brevity, repetition, vagueness, title-body overlap, or lexical patterns. However, it cannot prove that a review is fake. No ground-truth authenticity labels are available. For this reason, the results should be interpreted as content-based anomaly detection rather than fake-review classification. Future work could improve this module by incorporating account-level, temporal, or platform-level metadata, if available. Without such data, it is safer and more accurate to describe the output as textual unusualness rather than manipulation.

The fifth limitation concerns weather attribution. Weather was identified from review language, not from external meteorological records. This is appropriate for studying passenger attribution, because the key variable is whether the passenger mentions weather and uses it to explain the experience. However, it does not confirm the actual weather conditions affecting each flight. Future work could link reviews to historical weather data, airport-level conditions, and operational disruption records. This would allow the analysis to distinguish between actual weather exposure, passenger perception of weather, and airline communication about weather. The challenge is that the review data often provide only month-year flight information and incomplete route details, which limits flight-level matching.

The same issue applies to route, aircraft, and operational context. The dataset contains valuable contextual variables, but they are not always complete or standardised. Route strings may be inconsistent, aircraft mentions may be missing, and passenger descriptions may not identify the exact flight, aircraft registration, or disruption cause. As a result, contextual analyses are

necessarily approximate. Future work could connect review data to external aviation datasets, such as on-time performance records, aircraft schedules, airport congestion indicators, cancellation data, or route-level competition measures. This would allow the study of passenger perception to be compared directly with operational reality.

The sixth limitation concerns causal inference. The Difference-in-Differences design improves on simple before-and-after comparisons, but it still depends on assumptions. The most important is the parallel trends assumption: in the absence of the event, the treated period and the comparison period should have followed similar trends. This assumption cannot be fully proven with observational review data. It is more plausible for some shocks, such as COVID-19 or the Boeing Door Blowout, than for events shaped by broader demand or selection mechanisms, such as the Post-COVID Boom or EU Aviation Strikes. Future work could extend the causal framework using synthetic control methods, more granular time windows, airline-specific control groups, or external operational data. These extensions would make it possible to test event effects more rigorously.

The seasonal analysis also deserves further development. The thesis introduced a hemisphere-aware peak-season framework, which is more appropriate than treating seasonality as a single global calendar pattern. However, seasonality is more complex than a simple peak versus off-peak distinction. It varies by country, route, school calendar, holiday structure, airline network, and passenger purpose. Summer leisure travel, Christmas travel, pilgrimage travel, business travel cycles, and regional holiday periods may all create different forms of demand pressure. Future work could build a more granular seasonal model that accounts for origin and destination markets, route type, and traveller purpose. This would help identify whether the peak-season

penalty comes mainly from congestion, fare levels, staff pressure, aircraft utilisation, or passenger expectations.

Another limitation is that the thesis is intentionally broad. The pipeline includes eighteen modules and covers many dimensions of airline satisfaction. This breadth is valuable because it shows how ABSA can become a full analytical framework rather than a narrow classification task. However, it also means that some modules could be developed more deeply in future research. The weather module could become a full attribution study; the seasonal module could become a market-level resilience study; the causal module could become a dedicated event-impact analysis; the traveller-segment module could be connected to marketing personas; and the modelling module could be extended with gold labels, larger transformers, or multilingual representation. The thesis therefore provides a foundation rather than a final endpoint.

Future work should move in two complementary directions. The first is methodological: improve label quality, extend aspect extraction, validate the models against manual annotation, and integrate external data. The second is managerial: connect review-based insights with operational decision-making. Airlines could use a similar framework to monitor which aspects deteriorate during peak periods, which customer segments are most sensitive to disruption, whether weather communication reduces blame, and whether improvements in punctuality, recovery, or value perception translate into better satisfaction. In this sense, future research should not only ask whether the model is more accurate, but whether the resulting insight helps airlines make better decisions.

A further future direction would be to move from retrospective analysis to monitoring. The thesis analyses a historical corpus, but the same logic

could be applied to reviews as they are published. A live ABSA dashboard could track aspect sentiment over time, detect sudden increases in delay complaints, compare sentiment across routes or aircraft categories, and identify whether peak-season pressure is appearing earlier than expected. This would transform ABSA from a research tool into an early-warning system for customer experience. Such a system would be especially useful during disruption periods, when traditional operational metrics may not capture how passengers emotionally interpret the airline’s response.

Finally, future work could explore how review-based insight can be combined with direct managerial intervention. If ABSA shows that Value For Money is deteriorating on a specific route, airlines could test whether clearer fare communication, better service bundling, or targeted compensation improves perception. If delay-related language spikes during peak season, airlines could evaluate whether proactive communication reduces negative sentiment. If weather explanations are associated with more forgiving ratings only when they are specific, airlines could improve disruption messaging. In this way, future research could move from observation to experimentation, using ABSA not only to understand passenger satisfaction but also to test how it can be improved.

8.3 Final Remarks for Airline Management

The central managerial message of this thesis is that passenger satisfaction cannot be managed through the overall rating alone. A single score may reveal that a passenger was dissatisfied, but it does not explain why. It does not show whether the problem was a delay, the cabin product, the staff interaction, the price paid, the route context, the season, or the way the disruption was explained. For management, this distinction is critical.

General dissatisfaction is difficult to act on; aspect-level dissatisfaction is actionable.

The findings suggest that airlines should treat Value For Money as a strategic satisfaction driver. This does not simply mean lowering prices. It means managing the relationship between expectation and delivery. Passengers judge whether the experience was fair relative to what they paid and what they believed they were buying. If the fare is high, the service promise becomes higher. If the product is limited, expectations must be set clearly. If disruption occurs, recovery must be visible enough to protect the passenger's sense of fairness. Pricing, communication, product design, and service delivery therefore cannot be managed as separate functions. They all contribute to perceived value.

This has practical consequences for product strategy. Airlines often separate decisions about fare structure, cabin product, ancillary services, disruption recovery, and communication. Passengers do not experience them separately. A passenger who pays extra for a seat, baggage, lounge access, flexibility, or a premium cabin evaluates the entire journey against that perceived investment. If the service fails, the issue is not only the failure itself, but the perceived mismatch between payment and treatment. For this reason, Value For Money should be understood as a cross-functional measure. It belongs not only to pricing teams, but also to operations, customer service, revenue management, product design, and brand strategy.

The second managerial message is that punctuality and disruption management deserve special attention because delay behaves as a non-compensable damage factor. The results suggest that once delay becomes severe enough, satisfaction collapses toward a low level across traveller groups. This means

that airlines should intervene before frustration becomes irreversible. Communication should begin early, be specific, and continue throughout the disruption. Rebooking options, care, compensation, and explanation are not secondary services; they are part of the passenger's evaluation of the airline. When disruption is unavoidable, the way it is managed becomes the service.

This is especially important because passengers may forgive an unavoidable operational problem more easily than they forgive silence, confusion, or perceived indifference. A delay caused by weather, congestion, or safety requirements may be outside the airline's full control, but communication remains within the airline's control. The passenger may not expect the airline to control the weather, but they do expect the airline to explain the situation, protect vulnerable travellers, assist with connections, and provide a sense that someone is managing the problem. In this sense, disruption recovery is not simply a corrective action after service failure. It is a core part of service quality.

The third managerial message is that passengers interpret failures through attribution. Weather does not affect satisfaction only because it disrupts operations. It affects satisfaction because it changes how passengers assign responsibility. If passengers believe the cause is genuinely external, they may be more forgiving. If the explanation is vague, inconsistent, or poorly communicated, they may still blame the airline. This means that communication quality is not simply a customer-service detail. It is part of the causal story passengers build in their minds. Airlines should therefore explain disruptions with clarity, honesty, and operational detail when possible.

The fourth managerial message concerns peak season. The results show that satisfaction deteriorates during high-demand periods, especially in the Northern Hemisphere and especially for certain segments. This creates a

strategic challenge because peak season is also when many airlines carry more leisure passengers, families, and price-sensitive customers, often under conditions of congestion and reduced operational slack. For management, the lesson is that peak season cannot be treated only as a revenue opportunity. It is also a reputational risk period. Staffing, communication, boarding organisation, disruption recovery, and expectation management should be strengthened precisely when demand is highest.

Peak-season management should also be segment-specific. The results suggest that Premium Economy is especially exposed to satisfaction decline during peak periods. This is managerially important because Premium Economy is often positioned as a value bridge between Economy and Business. Passengers pay more than Economy passengers, but they may not receive the same level of protection, flexibility, or service recovery as Business passengers. If peak-season pressure reduces the visible difference between Economy and Premium Economy, passengers may feel that the price premium was not justified. This connects the seasonal findings back to Value For Money. The issue is not only that service worsens during peak periods; it is that the perceived fairness of the exchange may weaken.

The fifth managerial message is that customer experience should be monitored by aspect and context together. It is not enough to know that overall ratings declined in a given month. Managers need to know whether the decline came from Ground Service, Punctuality, Cabin Comfort, Value For Money, or communication during disruption. They also need to know whether the decline was concentrated in a specific carrier type, route length, cabin class, traveller group, weather condition, or peak-season period. This is where ABSA becomes valuable as a managerial tool. It connects what passengers say with where and when the problem occurs.

Finally, the thesis shows that airlines can use passenger reviews as a source of strategic intelligence if they analyse them at the right level of detail. Reviews are noisy, self-selected, and imperfect, but they contain patterns that are difficult to observe through numerical ratings alone. They reveal how passengers describe failures, which aspects they connect to dissatisfaction, where expectations are not being met, and how external conditions shape blame. A domain-aware ABSA framework does not replace operational metrics, surveys, or managerial judgement. It complements them by adding the passenger's own language.

The closing message is therefore straightforward: airlines should not listen only to whether passengers are satisfied, but to what passengers are satisfied or dissatisfied about, under which conditions, and compared with which expectations. The strongest improvements in customer experience will come from connecting operational reliability with perceived fairness. A punctual flight matters because it protects trust. A fair price matters because it protects the sense of exchange. Clear communication matters because it shapes responsibility. Service quality matters because it gives passengers evidence that the airline values them. When these elements are analysed together, passenger reviews become more than public opinion. They become a map of where management should act.

The broader implication for management is that customer experience should be treated as an integrated system rather than as a set of isolated service scores. The airline that improves punctuality but communicates poorly during disruption may still generate dissatisfaction. The airline that invests in cabin improvements but fails to manage perceived value may not receive the full benefit of that investment. The airline that explains weather honestly but does not provide recovery support may receive partial forgiveness but not

full satisfaction. The airline that treats peak season only as capacity utilisation may overlook the reputational damage caused by service pressure. The findings of this thesis therefore encourage an integrated approach: measure aspects separately, but manage the experience as a whole.

In conclusion, this thesis demonstrates that Aspect-Based Sentiment Analysis can help airlines understand passenger satisfaction in a more precise, contextual, and actionable way. The method reveals that satisfaction is shaped by fairness, reliability, expectations, attribution, and context. The results also show that the passenger's voice contains managerial knowledge that cannot be recovered from scores alone. The practical challenge for airlines is to turn that knowledge into action: protect reliability, communicate clearly, manage expectations, and design service recovery around the moments that matter most. If airlines can do this, passenger reviews become not merely a record of past dissatisfaction, but a guide for building better service in the future.

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